Breadth, Implied Volatility, and Forecasting with Stale Information

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Abstract

A forecaster holding a stale mark must buy something in place of timeliness, with no accepted way to price that trade. Handicapping an S&P 500 realised-variance forecaster by δ days, other sources current, we define the substitution rate $R(\delta)$, the share of delay damage a substitute repairs, and treat its inference as the weak-identification problem it is: the denominator is what the delay cost and can vanish, so every rate interval inverts a HAC t-test after Fieller at a fixed bandwidth fraction, with Kiefer–Vogelsang critical values, and is bounded exactly when that cost is itself significant. Unconditionally a current foreign cross-section returns 71% [54, 83] at eleven weeks – by which point the delay has taken all but a hundredth of the undelayed model's skill, so the figure is that model's skill recovered rather than a share of anything still standing. Conditioning on a VIX tercile, the level taken at $t-1$ and the boundaries fixed on the whole window, gives 81% [67, 90] under stress and 53% [28, 76] in the middle, with calm markets too weakly identified to support an estimate at all; the three are reported as description and not as a difference, because their confidence sets overlap. What the design does separate is the share of domestic error the cross-section removes, 42.0 points [25.2, 55.8] higher under stress. Given own-index implied volatility the cross-section adds nothing detectable on five of the six markets examined and significantly subtracts on the sixth; a chain on another asset does not do the same. Twenty house price indices, fourteen of them with full histories and carrying the balanced inference, carry the measurement to a smoothed market with no listed options chain of its own. That is an appraisal-like published index, not an appraised book.

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JEL codes: C53, C58, G17, C22.

1. Introduction

A private equity portfolio is marked once a quarter. Direct property is appraised on a cycle, and the resulting series is smoothed in ways that have been documented for thirty years (Geltner et al. 2003). Hedge fund returns arrive with a serial correlation that understates their true volatility (Getmansky et al. 2004). In each case someone must produce a risk number today from marks that are weeks or months old, while public markets have gone on trading in plain sight.

Two public markets are candidates. The first is breadth: other markets are correlated with yours, and some of them have already closed today. The second is the options market, which reports a forward-looking risk number continuously and without any delay at all. These are different purchases, and a practitioner choosing between a wider data feed and an options subscription would like to know which one buys back more of what staleness took.

We cannot observe the truth for a private portfolio, so we run the experiment where the truth is observable. We take a forecaster of S&P 500 realised variance, withhold their own data for δ days,

leave the other sources untouched, and measure what each gives back. For foreign equities the asymmetry costs nothing to construct: Sydney and Tokyo close at 06:00 UTC, Hong Kong at 08:00, Mumbai at 10:00, London and Frankfurt at 16:30, New York at 21:00. Six of the seven markets used here have therefore finished trading before New York does; the seventh, São Paulo, closes at the same hour and contributes its previous session, which is the rule Section 3 states and applies without exception.

An experiment run where the truth is observable is still a proxy for the case that motivates it, and a paper that stopped there would be asking its reader to extrapolate from liquid equities to appraised assets and to trust the extrapolation. The last step is therefore to narrow that extrapolation as far as public data allows, with a housing-index panel that has appraisal-like persistence and infrequent marking. Twenty metropolitan house price indices, fourteen of them published monthly since January 1987, carry the properties the first paragraph describes, though they remain a published index and not a book. Each monthly figure is a three-month moving average of closings; no options chain on any of the twenty is currently listed and no daily own-market implied-volatility index exists for them, so the substitute Section 7 races against is unavailable here; CME does list futures on the composite and on ten of the metro indices, which is a derivative but not the one that race needs. All twenty are published on the same day, so a stale mark sits inside a fresh cross-section exactly as the design requires. The same measurement is then made there, on the same estimator.

Four findings carry it. First, breadth substitutes for timeliness and the amount can be measured on one ruler. The amount is reported where it is determined rather than where it is largest. A current foreign cross-section returns 81% [67, 90] in this sample's stressed markets, cut on a VIX tercile whose boundaries are fixed on the whole window, which is a description of where the mechanism was strong and not a rule a forecaster could have applied on the day. In calm markets the same quantity spans more than a hundred points and has no usable value, for the reason Sections 4.5 and 4.6 give. Between the two sits a middle tercile at 53% [28, 76], lower at the point estimate but not separated from the stressed one: under the construction of Section 4.6 the three states' confidence sets overlap at every delay. The ordering itself is graded on the share of domestic error the cross-section removes, where it can be tested as a difference, and not on the rate, whose own contrast spans zero; Section 5 gives both.

Pooling the three, a current foreign cross-section returns 71% [54, 83] on the implementable target, with the original model, the stronger control, the compressed specifications and the state-space alternative all landing between 67.4% and 75.3%, all on the same test days, with the same estimator and a nearly identical information set, so their agreement is close to mechanical and is not independent corroboration of the figure. The interval prices sampling noise and nothing else, and the choice of control is priced separately rather than left implicit: across the eight domestic controls of Section S4 the rate at eleven weeks runs from 60.7% to 74.3%, a spread of 14 points against the twenty-nine the interval spans, under a selection rule fixed before any rate was read. At eleven weeks the delay has taken all but a hundredth of the undelayed model's skill, so that figure is that skill recovered rather than a share of anything still standing; Section 4.5 says which reading each part of the delay grid supports. Table 1 collects them beside the question each one answers. Read as a clock rather than a share, a fifty-five-day-old mark plus today's foreign close forecasts as well as a mark 4.6 days old [2.6, 8.2].

Second, an options market on the asset itself leaves the cross-section no detectable increment: redundant in the bounded sense of Section 7, which rules out a gross gain above 0.022 of R^2 rather than establishing equality. That holds only on the asset itself. Where the only chain available is written on another index the two candidates finish three points apart rather than thirty-one, and a compressed cross-section adds in thirteen of twenty-four cells where the seven-regressor block added in one. The compression was adopted after both counts were in hand and few of the thirteen survive a post-selection correction, so the evidence supports the pattern and not any single cell. That phrase is load-bearing and has a test behind it: the exploratory grids hold more structure than the jointly resampled null produces. Section 9 gives the test and Section S37 the working.

Third, which of those worlds a holder is in can be measured before anything is bought, because the target's own correlation with the cross-section orders the substitution rate across the six markets examined, at a rank correlation of +0.943 whose worst leave-one-out value is +0.900; what six points in one cluster do not support is the fitted line itself, nor any statement about a weakly coupled market, since every target measured here couples between 0.612 and 0.808. Table 5 grades the ordering and the line separately, on a four-grade scale whose terms are this paper's own and carry no meaning beyond it.

Fourth, on the house price panel one quarterly appraisal cycle of extra staleness gives 18.8% [6, 31] on the variance coordinate, positive on thirteen metros of fourteen. That figure is the output of a peer-block choice, which has to be stated: three candidate blocks were priced, and on this coordinate the widest makes the rate *negative* at one, two and three months, at −88.7%, −30.8% and −1.9%, while the compressed block is positive at every delay. Thirteen peer columns cost 0.0396 of R^2 before any substitution is available to pay for them. The block is chosen by a rule that cannot see the rate and breaks ties towards fewer regressors, and Table S27 of the Internet Appendix prices all three. At this sample size the choice is the difference between the claim holding and appearing to fail, so it is reported here rather than in the appendix. That coordinate is the one this paper's target is defined on, and the harder of the two: forecasting a trailing twelve-month variance means forecasting something that overlaps its own current window, so the denominator it divides by is small. On next month's return, which is the quantity an owner cares about but is not this paper's target, the same design gives 67.9% [43, 91], positive on fourteen of fourteen.

Table 1. The substitution rate at eleven weeks under each specification the paper regards as defensible, with the question each one answers. All are R²-scale. The headline row is the implementable origin-dated figure of Section 4.2; the four comparison rows are on the target-dated convention the sections that compute them use, which is why the headline here is 71% and Table 2 reads 71% as well. At eleven weeks the two conventions agree to a third of a point and at shorter delays differ by at most 1.1, so the spread below is read as a band and not as six figures on one ruler. The first five are on the same 4,862 test days as Section 5; the stressed row is the 1,618 of those days in the upper volatility tercile; and the last is a different experiment on a different asset class at a different delay, read on the three non-overlapping quarterly phases of the 311-month panel rather than month by month, included because it is the result the paper is built to deliver. It is not the monthly headline, which is 67.9% on the return coordinate and 18.8% on the variance coordinate; Section 8 separates the three. No row here uses the 4,525-day window of Section 7, which is shortened by VDAX-NEW availability and carries the options race alone. Every interval here is the fixed-b Fieller set of Section 4.6, at the bandwidth and critical value stated there, so the rows are comparable to each other and to the abstract. The bootstrap equivalents are in the replication output and are narrower, which is the direction Section 4.6 measures. The three market-state rows are the same specification cut on a VIX tercile, the level at t−1 and the boundaries fixed on the whole test window, and the calm row carries no figure because its confidence sets are too wide to support an informative estimate: where delay costs a forecaster almost nothing, the share of that cost repaired is a ratio of two small numbers. That is imprecision rather than demonstrated weak identification; Section 5 gives the denominator statistics. Two rows carry none, because their evidence is a paired difference rather than a marginal interval: the stronger control is 4.1 points below the paper's own, interval $[-10.6, -0.3]$ on that difference (Section S4), and the state-space filter is 3.8 points above it with its own parameters frozen at the training block, 72.6% when they are re-estimated as the window grows (Section S20).

specification	R(55)	the question it answers
implementable target, original block	71% [54, 83]	what a forecaster could have acted on
strongest domestic control	67.4%	is it an artefact of a weak benchmark?
equal-weighted cross-section	74.3% [59.1, 84.1]	what a practitioner should actually run
one real-time factor	75.2% [60.1, 84.9]	where in the block the recovery lives
state-space alternative, parameters frozen	75.3%	does a better estimator change it?
stressed markets only	81% [67, 90]	when is the mechanism worth having?
middle tercile only	53% [28, 76]	and when is it worth less?
calm markets only	not determined	where does the question stop having an answer?
house prices, one quarter, non-overlapping	*57%*	*does it hold on a smoothed index of an illiquid market?*

Nothing turns on which row is called the headline: the span is narrow relative to the intervals the table carries. The 71% is the lower choice, being the implementable target and the original specification, while the 74% to 75% band is what the compression of Section 6.1 recommends; the paper quotes the first and recommends the second.

The contribution is four-fold, and the first is a measurement problem rather than a finding. What a stale mark costs and what a substitute returns are both differences in forecast skill, so the quantity a buyer wants is their ratio. This paper defines it as $R(\delta)$ and gives it a real-time design in which the training window stops $\delta + h$ days before the forecast origin, so that no label used to fit the model had resolved after the stale mark was current.

Unless a figure says otherwise, an interval quoted above is a fixed-b Fieller set, obtained by inverting a HAC t-test after Fieller (1954): a construction free to return a bounded interval, two half-lines or the whole line, which is what a ratio whose denominator can vanish requires. Three families are excepted

and are named where they appear: the effective age, the house price figures and any difference between two rates. A moving-block bootstrap of the same ratio (Künsch 1989) is kept as an implementation check. The two do not agree, and the disagreement is not a tie to be split: the inversion is wider at every delay, and the calibrated generator separates them: the inversion is close to nominal, while the bootstrap is short where the denominator is well identified and over-covers at several times the width where it is not. Section 4.6 gives the calibration, the critical value, the taper and why the apparent agreement of the two procedures under normal critical values is not corroboration. The measure and that treatment of it are what the rest of the paper reuses, across loss functions, markets, candidate substitutes and one asset class outside equities.

The other three follow from applying it. For a fixed loss the rate is scale-free and comparable across targets and horizons, so a wider data feed and an options subscription go on one ruler rather than being argued about; across different losses it indexes distinct estimands, and Section 4.5 says what the three reported here do and do not share. The scope condition is identified in the examined markets rather than assumed: it turns on whether an option chain exists on the asset itself, and on a coupling statistic computable before anything is bought. And the illiquid case, which this literature has treated as the motivating example and left unmeasured, is brought within one step of measurement. One step, not none.

The paper proceeds as follows. Section 2 places it against what is already known. Sections 3 and 4 give the data, the design, the estimator, the loss functions, the definition of $R(\delta)$ and its inference. Sections 5 to 7 measure what breadth recovers, where it stops and how it fares against implied volatility; Section 8 carries the measurement to house prices; Section 9 tests it and bounds where it holds. Section 10 says what the paper cannot say and Section 11 what would test it next. The Internet Appendix carries the working, the robustness set and the practitioner's reading.

2. Related work

Four findings underpin this exercise, and none of them is new here.

Foreign equity volatility forecasts domestic volatility. Engle, Ito and Lin (1990) rejected the idea that volatility is a purely local phenomenon, and the spillover literature since has measured the connections in considerable detail (Corsi 2009; Diebold and Yilmaz 2014). Korkusuz augments a HAR model of S&P 500 volatility with a European block and improves on it (2025). Buncic and Gisler (2016) run the mirror experiment, forecasting seventeen foreign markets from US volatility, and report out-of-sample R^2 between 4.6% and 14.5%.

Non-synchronous closing times define which observations a forecaster may legitimately use. Zhao and Qiao say so explicitly and build an attention model whose mask prevents a closed exchange from acting as a source (Zhao and Qiao 2026). Jayawardena and co-authors forecast Tokyo volatility from markets that trade while Tokyo sleeps (2020).

A live market can price a stale one. Goetzmann, Ivković and Rouwenhorst showed that same-day US returns predict the next day's international fund NAVs strongly enough to trade on (Goetzmann et al. 2001): the mechanism of this paper stripped to its simplest form.

Implied volatility carries much of the cross-market gain. This is the finding that shapes the design below. Both Korkusuz (2025) and Buncic and Gisler (2016) report that a large share of what looks like a cross-market effect is attributable to implied volatility, the VIX in particular. Their work does

not settle the matter for a *delayed* forecaster, since in both cases the domestic information set is current, but it makes the comparison unavoidable: any claim that foreign markets substitute for timeliness has to be set against an options market that is current by construction.

A second literature has the same problem and a different name for it. Macroeconomic nowcasting is built around exactly the asymmetry this paper constructs on purpose: series arrive on different schedules, so the most recent rows of a panel are present for some variables and missing for others. Wallis called it the ragged edge (1986), and the modern treatment is a dynamic factor model with a Kalman filter that handles the missing rows in the measurement equation rather than by imputation, which is how Giannone, Reichlin and Small measure the value of each release as it lands (2008) and how Doz, Giannone and Reichlin estimate the factor in two steps (2011). The mixed-frequency alternative regresses a low-frequency target directly on high-frequency data with a parsimonious lag polynomial, which is MIDAS (Ghysels et al. 2007), and the two approaches have been compared on precisely the ragged-edge problem (Marcellino and Schumacher 2010). A third strand insists on the vintages: Croushore and Stark's real-time data set exists because a forecast evaluated on revised data is not the forecast anyone could have made (Croushore and Stark 2001). And forecast combination is the standing answer to not knowing which of several models to trust (Timmermann 2006).

The connection is close, and the differences are what leave room for this paper. The apparatus is shared: our design is a ragged edge with one variable's edge pushed back by hand, our Kalman comparison in Section 10 is a nowcasting model, and our compressed peer block is a factor. What is not shared is the question. That literature asks how accurately the latent current value can be estimated, and answers in the units of the target. We ask what share of a *known* loss a *named purchasable* source repairs, and answer, for a fixed loss, in a scale-free ratio that lets a data feed and an options subscription be compared with each other. Nor is the ragged edge in that work a variable: it is whatever the release calendar happens to deliver, so the delay is a fact about the data rather than a parameter to be swept. Sweeping it is what produces a delay curve, a ceiling and a rate.

The stale-price literature in finance is the other half of the inheritance and is older. Scholes and Williams (1977) and Lo and MacKinlay (1990) showed what non-synchronous trading does to measured covariances, which is why a cross-section of closes taken at different hours cannot be treated as simultaneous; Getmansky, Lo and Makarov (2004) and Geltner and co-authors (2003) showed what smoothed or infrequent marking does to measured risk. Closest of all in spirit is Brown, Ghysels and Gredil (2023), who nowcast private equity net asset values from public market information between reporting dates: the same act on the same kind of asset, reported as a level rather than as a price paid for timeliness. Section 8's house price panel is the object that literature and this one share, and the substitution rate is what this paper adds to it.

What none of this work does is treat the delay as a variable. In every case additional data is added to an information set that is already current, and the question is whether it helps. The complementary question, and this paper's, is: when domestic information is not current, how much of the damage does each candidate undo, and how does that fraction move as the damage grows? That is a rate of substitution between things a practitioner can actually buy, and we could not find it measured.

3. Data

The data are daily open, high, low and close for eight equity indices: S&P 500 (SPX), Nikkei 225 (N225), S&P/ASX 200 (AXJO), Hang Seng (HSI), Nifty 50 (NSEI), FTSE 100 (FTSE), DAX (DAX)

and Bovespa (BVSP), from January 2000 to September 2026 as raw data, of which the out-of-sample evaluation uses 4,862 days once the burn-in, training and validation blocks of Section 4.3 are spent, together with eight implied-volatility indices: VIX, VIX9D and VDAX-NEW, which carry the main design; own-index indices for Tokyo, Mumbai, Hong Kong and Sydney, used only in Section S40; and VSTOXX, which Section S33 tests and keeps out. The two CBOE series are Cboe's own published histories, Sydney's own-index series is from Yahoo Finance for the reason Section S40 gives, and the rest are retail end-of-day exports from Investing.com, downloaded in September 2026. The repository lists every file, the identifier it maps to, and the column dictionary, and pins the environment to NumPy and pandas alone.

Section S14 of the Internet Appendix adds two further index series, for Indonesia and Malaysia, from a different vendor and as one file each rather than two. They are kept in their own folder, and what rests on them is stated here: they supply two of the six targets behind the coupling result of Section 9, and they are the two least coupled of the six. They carry less verification than the primary exports, because arriving as single files they have no second export to check against, so the audit below does not reach them. Two further series from the same vendor, for Pakistan and Sri Lanka, were used in an earlier vintage of this study and have since been withdrawn by it. They cannot now be obtained, by a reader or by us; nothing in this paper is computed from them, and the data availability statement records that.

Each retail series came in two exports, because the source caps a download at 5,000 rows, which leaves nine joins to be checked. Eight of the nine overlap by 120 to 251 trading days, and across every overlapping row the two exports agree exactly: no mismatches, largest absolute difference zero. The ninth is the S&P itself, where the exports abut without overlapping and no cross-check is possible. We report it as unverified and fall back to two weaker tests: the join is contiguous, Friday 9 March 2018 to Monday 12 March 2018 with no session missing or duplicated, and the return across it is -0.0013, or 0.11 standard deviations of the series' own daily returns, so there is no level shift where the exports meet.

A foreign close is admissible only when its timestamp falls before the target's. Markets closing at or after the target therefore contribute their previous session. The rule generalises without special cases: a Tokyo forecaster gets every other market a day late, because Tokyo closes first (Section S9). It is implemented from a table of standard-time closes, and whether that simplification is safe is a question about real exchange clocks rather than about the design, so Section S35 rebuilds every close in UTC from its exchange's own local closing time and timezone, daylight-saving transitions included, on all 6,991 dates. Across every target, every peer and every date there are **zero** days on which the true clocks would have reordered a block, and the narrowest margin the rule relies on anywhere is two hours. The audit also shows the implementation is stricter than the rule it implements, which is the direction that costs information rather than validity; Section S35 quantifies it. It also disciplines the implied-volatility block. VDAX-NEW satisfies the rule outright, since Frankfurt finishes hours before New York. VIX does not. Its printed daily close is stamped 21:15 UTC, fifteen minutes after the 21:00 S&P cash close, so that print is inadmissible. VIX is computed continuously from live SPX option quotes, so a value did exist at 21:00 and would be the right observation, but a retail end-of-day export carries no intraday snapshot. We therefore run the two specifications either side of it: the previous day's close, strictly admissible and discarding a full day of information, and the same-day close, which contains fifteen minutes we are not entitled to. Section 7 leads with the lagged arm, because it is the one that obeys our own rule, and reports the same-day

arm as a sensitivity check. The 21:00 observation is temporally between the two, which does not imply that its forecasting performance lies between theirs. Timestamps order; skill need not.

4. Methodology

The object of this section is one number and the conditions under which it means anything. A forecaster loses skill when their own mark goes stale, and a substitute returns some of it; the quantity a buyer wants is the second as a share of the first, which is the ratio $R(\delta)$ of equation (9). Everything before that equation exists to make the two skills comparable: one volatility estimator, one decision variable, one real-time training window that stops $\delta + h$ days before the forecast origin, one family of losses. Section 4.6 is what stops the ratio being read further than the data supports, since its denominator is what delay costs and approaches zero wherever delay costs little.

4.1 The volatility estimator

Volatility is the Yang-Zhang estimator (2000) over rolling five-day windows, which uses the whole daily range and so sees more of each session than the close alone. Write O_s, H_s, L_s and C_s for the open, high, low and close of session s, and

$$o_s = \log(O_s / C_{s-1}), \quad c_s = \log(C_s / O_s), \quad u_s = \log(H_s / O_s), \quad d_s = \log(L_s / O_s)$$

for the overnight, open-to-close, high and low log returns. Over the n sessions ending at t the estimator is

$$\sigma^2_{YZ,t} = \sigma^2_o + k\,\sigma^2_c + (1 - k)\,\sigma^2_{RS}, \quad k = 0.34 / (1.34 + (n+1)/(n-1)), \tag{1}$$

where σ^2_o and σ^2_c are the sample variances of o_s and c_s over those n sessions, σ^2_{RS} is the mean over them of the Rogers-Satchell term $u_s(u_s - c_s) + d_s(d_s - c_s)$ (Rogers and Satchell 1991), and $n = 5$ throughout. The three components measure the overnight gap, the drift within the session, and the range net of that drift. The sum is drift-independent under a Brownian price for *any* k, because the Rogers-Satchell term is itself drift-independent (Rogers and Satchell 1991) and the other two are demeaned; what k buys is variance, and the value in (1) is the weight at which the combination is minimum-variance (Yang and Zhang 2000). It rises with n, because the demeaned open-to-close estimator spends a degree of freedom on the mean and so is relatively worse in short windows, which puts the five-day window used here at the low end of its range. All variances use the $n - 1$ divisor.

Why this estimator, and what turns on it. Every figure in this paper is computed on this one measurement of the object being forecast, and one feature of it must be dealt with first. The overnight term is built from the gap between one New York close and the next New York open, exactly when the foreign markets used as the substitute trade. The domestic feature therefore contains, by construction, a reading of the same global factor the cross-section carries, and the design cannot by itself separate breadth substituting for a stale mark from the overnight component of the target being a non-synchronous average of those same sessions. Section S38 rebuilds the measurement under four proxies: the rate at eleven weeks reads 71.5% on the published estimator, 69.4% and 70.1% on two intraday estimators that contain no overnight term at all, and 73.4% on close-to-close, which uses no open, high or low. The component the alternative explanation needs can be removed entirely for about two points of the rate, so it does not survive. What this does not establish is that the target is

well measured: no intraday data is used anywhere here, and a realised-variance target would be a better object than any of the four.

4.2 The decision variable and the forecast target

Let i index the eight series and t the target's trading day. The decision variable each model sees is the log of that series' variance against its own recent level,

$$a_{i,t} = \log(\, \sigma^2_{YZ,i,t} \,/\, M_{i,t} \,), \quad M_{i,t} = \text{median}(\, \sigma^2_{YZ,i,s} : t - 251 \le s \le t \,), \tag{2}$$

and the quantity to be forecast, with horizon h = 5 trading days and target market denoted SPX, is

$$y_t = a_{SPX,\, t+h}. \tag{3}$$

So $M_{i,t}$ is the trailing 252-day median, and the implied-volatility series enter the same way, as the log of the level over its own trailing median. Dividing by a series' own median puts Tokyo, São Paulo and the VIX on one scale and keeps each series' units out of the regression. The target at t + h carries a median whose window includes the five days between origin and outcome, which a forecaster at t does not know; no feature is affected, since every one is dated $t - \delta$ or earlier for the domestic block and t for the others. Section S16 runs the whole experiment under the implementable alternative, which normalises the outcome at the origin.

One definition per place, stated once: unless a caption says otherwise, the equity-result tables in this paper report the target-dated rate and every headline sentence reports the implementable origin-dated one. Table 1 is the labelled exception, since it collects the headline specifications and therefore carries the origin-dated figure, and its housing row is a different experiment on a different asset class. The two differ by at most 1.1 points at every delay, and Table S12 gives both columns. The target is a_{t+5}, five trading days ahead, left continuous: a companion note (Vellanikaran 2026) showed that splitting such a target at its median throws away exactly the magnitude information a cross-sectional model would use, so we do not split it.

4.3 Feature blocks and estimation

Models are built from three feature blocks. The domestic block is the target's own history at daily, weekly and monthly aggregation, in the manner of Corsi (2009), observed at $t - \delta$. The foreign block is seven index decision variables at their admissible timestamps: the six that close before New York, plus Bovespa a session late. The implied-volatility block is VIX and VDAX. Only the domestic block is delayed; the others are what a forecaster could see today, and that asymmetry is what the comparison isolates. Every model is a ridge regression refitted every 21 days, with features standardised on training statistics alone. Formally, let x_t be the feature vector a model sees at t, assembled from the blocks above. Training cannot run up to t: the most recent pair whose outcome had resolved by the time the target's own mark was current ends at

$$c_t = t - \delta - h, \quad T_\delta(t) = \{c_t - 1499, \ldots, c_t\}, \tag{4}$$

so the window holds 1,500 days and slides with δ as well as with t. δ does not only make one number old. It freezes the whole domestic information system for δ days: the current state, the labels available for fitting, the coefficient vintage, the ridge penalty the validation tail selects, and the quantity of training data. What this paper measures is therefore a domestic-data *outage* of δ days, not a stale mark beside an otherwise current desk, and $R(\delta)$ is the share of an outage's damage that a substitute

repairs. Section S36 prices the difference by running a third arm that no forecaster could run, with a stale input but coefficients refitted to $t - h$, and finds that freezing the estimation accounts for 0.4% of the damage at a week and 2.7% at eleven weeks. The denominator $R(\delta)$ divides by is information rather than estimation, and the rate computed inside the infeasible arm is 2.8 points higher at most, so the paper's design is also the conservative one. Features are centred and scaled by that window's own mean μ_T and standard deviation s_T, giving $z_t = (x_t - \mu_T) / s_T$ elementwise, and the coefficients (β_0, β) solve

$$\min_{\beta_0, \beta} \sum_{u \in T_\delta(t)} (y_u - \beta_0 - z_u'\beta)^2 + \lambda \beta'\beta, \tag{5}$$

with the intercept β_0 left unpenalised. λ is chosen from $\{0.3, 1, 3, 10, 30\}$ in two steps: candidate penalties are fitted on the first 1,250 days of $T_\delta(t)$ and scored on the last 250, which are held out of the fits that select between them, and the chosen λ is then refitted on all 1,500. The validation tail is therefore held out of the candidate fits, which is what lets it choose between them, and is not held out of the final model. The forecast is $f_t = \beta_0 + z_t'\beta$. Standardising on training statistics alone matters: scaling a feature by a full-sample standard deviation would let the test period influence the fit through the scale, which is a small leak but a real one. All are trained on the same labels, a pair being admitted only once its outcome was resolvable at $t - \delta$, so differences between them are features and nothing else. The rolling fixed window is not incidental; it is what makes the tests in Sections 5 and 6 valid. Figure 1 draws the arrangement, in which six dates matter.

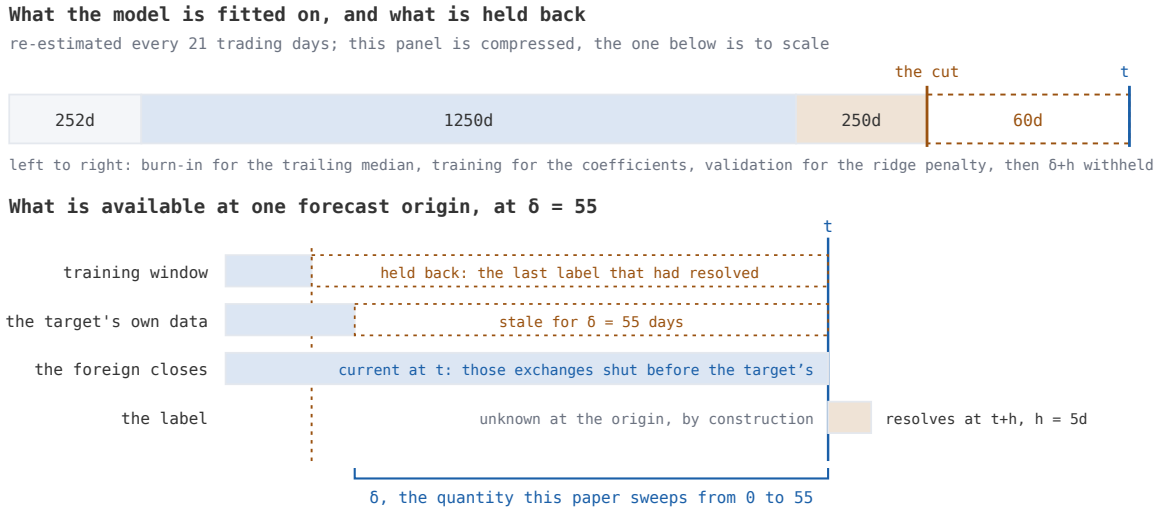


Figure 1. The design of Section 4, on two panels. Above, the window every model is fitted on, drawn on a compressed axis: a burn-in the trailing median of equation (2) needs before it exists, 1,250 days on which candidate ridge penalties are fitted, a 250-day tail on which they are scored, and then the δ+h days withheld because their labels had not resolved. The chosen penalty is refitted on all 1,500, so the tail is held out of the candidates' own fitting, is what selects among them, and then enters the final model. Below, one forecast origin at the delay this paper leads with, drawn to scale. At t the training window stopped δ+h days ago, the target's own mark is δ days old, and the foreign closes are current, because those exchanges shut before New York does. The cut at $t - \delta - h$ is a purge in the cross-validation sense and is load-bearing: Section S7 measures what removing it inflates.

4.4 Loss functions and proxy robustness

With an imperfect volatility proxy only a restricted family of losses preserves the ranking a researcher would obtain against true latent volatility (Patton 2011), and that family is defined on the VARIANCE scale. Our target is a log variance, so the two halves of the usual pairing land differently.

QLIKE survives the transformation in the sense that matters for the ranking: it depends on the ratio of outcome to forecast alone, and multiplying both by the same positive number leaves that ratio untouched, so it may be computed on the variance scale or on any common rescaling of it. That invariance is not a claim that the forecast it scores is free of the reconstruction below: outcome and forecast carry different medians, and the smearing factor is in the forecast. Squared error does not. Computed on log variance it is outside the family, because a monotone transform of the arguments is precisely what the robustness result fails to survive.

We therefore report three, and state the reconstruction back onto the variance scale in full: an error in it is how a loss comes to be called robust when it is not. The benchmark forecast is not a single number. At each test day t it is b_t, the mean of the labels a forecaster had already seen, over the 1,500 days ending at $t - h$, where labels stop resolving for someone whose own data is current. Two properties are load-bearing. It is not delay-matched: one benchmark serves every model at every delay, because $R(\delta)$ divides by $S_{own}(0) - S_{own}(\delta)$, and a yardstick that moved with δ would mix the damage the delay does with the change in the yardstick. And it is a forecast, not a fitted constant: a mean taken over the test period would be chosen with hindsight, which is not what out-of-sample skill means. Write n for the number of test days. Realised variance is recovered from the target by undoing the normalisation at the date the target carries it, $v_t = \exp(y_t)\, M_{t+h} = \sigma^2_{YZ,t+h}$. A forecaster at t does not know M_{t+h}, so the matching forecast can only use the median they do have, $w_t = \gamma_t \exp(f_t)\, M_t$, which carries the median's h-day drift as forecast error rather than hiding it. The three losses are

$$R^2 = 1 - \sum_t (y_t - f_t)^2 \div \sum_t (y_t - b_t)^2, \tag{6}$$

$$\text{QLIKE} = (1/n) \sum_t [\, v_t/w_t - \log(v_t/w_t) - 1 \,], \tag{7}$$

$$\text{MSE}_{var} = (1/n) \sum_t (v_t - w_t)^2, \quad \gamma_t = (1 / |T_\delta(t)|) \sum_{u \in T_\delta(t)} \exp(y_u - f_u), \tag{8}$$

where γ_t is Duan's smearing factor (1983), estimated from the same delayed training window at each refit. One factor per refit tracks drift in the residual scale over time, but it is constant across the regressors within a window, and a single factor is conditionally unbiased only where $E[\exp(y - f) \mid z]$ does not depend on z. Where the log-scale residual scale varies with the predictors it does not, and the level forecast is then biased in the tails of z in a way a window average cannot see, materially so when the scale varies strongly with the predictors.

The correction is applied to the benchmark on the same schedule, so a skill score differences most of it away. R^2 is computed on the log scale and needs no retransformation at all. QLIKE and MSE_{var} are both computed from w_t, so the smearing factor enters both and not the variance-scale column alone. We report that as a limitation of the two retransformed losses rather than repairing it, because a regressor-dependent smearing estimate would add a second fitted object to the losses we lean on least; MSE_{var} is the more exposed of the two, because squared error on the variance scale punishes the gap between a median-unbiased and a mean-unbiased level directly, while QLIKE sees it only through the ratio. Skill under QLIKE and MSE_{var} is defined as in (6), one minus the model's loss over the benchmark's, so all three columns are shares of a benchmark loss removed and are read the same way.

The benchmark is carried onto each scale the way a model forecast is: exponentiate, apply its own smearing factor γ^{b}_{t} estimated on the same resolved labels, and multiply by the median the forecaster holds, so the benchmark's natural-scale forecast is $\gamma^{b}_{t} \exp(b_t)\ M_t = M_t \cdot \text{mean}_u \exp(y_u)$. Since y is a normalised log variance, $\text{mean}_u \exp(y_u)$ is the trailing mean of variance over its own median and not the trailing mean of variance: on this panel the two differ by 62.8% of the mean level and correlate 0.35. Section S25 scores a benchmark built directly from resolved natural variances beside the reconstructed one, so the choice is visible. The rate is identical under either, to the decimal place, because the benchmark cancels from a ratio of skill differences. Leaving the benchmark unsmeared while the models are smeared would deny it a correction they receive and overstate skill on both variance-scale columns; giving it the median at t + h while the model only holds the median at t would do the reverse, and the second is the error this reconstruction exists to avoid.

Of the three, R^2 in (6) is the paper's headline and is *not* proxy-robust; QLIKE in (7) is, and Part 0 of the script behind this section shows it exactly rather than by argument, computing QLIKE from v_t and w_t and again from both divided by the same median and getting the same number to ten decimal places – which establishes that QLIKE is unchanged by a common rescaling of outcome and forecast, and not that w_t is free of the retransformation; and MSE_{var} in (8) is Patton's MSE on the variance scale and is also robust. Computing MSE_{var} on the median-normalised scale instead, while calling it Patton's MSE, is the error this definition forecloses; Section S25 reports what it costs, which reaches 29 points of the rate. Every MSE_{var} figure in this paper is the natural-scale one defined above; Section S25 reports both columns and the gap between them. The smearing factor is there because exp() of a log forecast is median-unbiased, not mean-unbiased, and squared error on the variance scale punishes exactly that, so the reconstruction needs the correction, not the raw exponential. Where conclusions differ across losses we give weight to what both proxy-robust losses support; where one of them cannot resolve a row, we rest no claim on it and say so where that happens.

4.5 The substitution rate

Write $S(\delta)$ for skill at delay δ under any one of (6), (7) or (8): the share of the benchmark's loss that a model removes. $S_{\text{own}}(\delta)$ is the skill of the model holding only the domestic block, δ days stale; $S_{\text{alt}}(\delta)$ is the skill of the same model with a candidate substitute added. Delay costs $S_{\text{own}}(0) - S_{\text{own}}(\delta)$; the substitute returns $S_{\text{alt}}(\delta) - S_{\text{own}}(\delta)$. The substitution rate is their ratio,

$$R(\delta) = [\ S_{\text{alt}}(\delta) - S_{\text{own}}(\delta)\ ] \div [\ S_{\text{own}}(0) - S_{\text{own}}(\delta)\ ]. \tag{9}$$

What (9) estimates. Equation (9) is written in sample skills, which is how a practitioner computes it, and the inference of Section 4.6 is inference about a population quantity, so the two need connecting rather than leaving adjacent. Let $L_{m,t}(\delta)$ be model m's loss on test day t and define the two per-day differences

$$d_{1,t} = L_{\text{own},t}(\delta) - L_{\text{alt},t}(\delta), \quad d_{2,t} = L_{\text{own},t}(\delta) - L_{\text{own},t}(0). \tag{10}$$

Because every skill in (9) divides the same benchmark loss, the benchmark cancels from the ratio exactly. Writing $Q(f) = \sum_t (y_t - f_t)^2$ for a forecast sequence f and $B = Q(b)$ for the benchmark's own loss, (6) makes $S(f) = 1 - Q(f)/B$, so a difference of two skills is $[Q(f_2) - Q(f_1)]/B$ with the additive constant already gone; numerator and denominator of (9) therefore both carry $1/B$, which cancels,

leaving a ratio of two sums of the per-day differences in (10). Nothing in the rate depends on which benchmark was chosen: the fixed benchmark of Section 4.4 is needed for reading $S(\delta)$ as a curve, not for $R(\delta)$. So (9) is the plug-in

$$R(\delta) = \text{mean}_t(d_{1,t}) \div \text{mean}_t(d_{2,t}) \quad \text{for} \quad R^*(\delta) = E[d_{1,t}] \div E[d_{2,t}]. \tag{11}$$

$R^*(\delta)$ is the estimand: the share of the expected loss a delay imposes that a substitute is expected to return. Under stationarity and ergodicity of the loss-differential process, the plug-in converges to it as the evaluation length n grows, provided $E[d_{2,t}] \neq 0$, which is the same condition Section 4.6 is about, seen from the other side: where the denominator's expectation is zero the estimand does not exist, and where it is near zero the estimand exists but no procedure that is uniformly valid over a parameter space admitting weak identification can always return a bounded set (Dufour 1997). A bounded interval can still be valid pointwise, at a denominator separated from zero; what cannot be had is that guarantee uniformly. Stationarity over a nineteen-year test window is an assumption rather than a fact, and Section S5 reports the rate on each half of the sample separately for that reason. Everything in Sections 4.6 and 5 is inference about $R^*(\delta)$; everything in the tables is the plug-in.

It has no units, so for a given loss it can be compared across markets and across candidates whatever their scales. Across *different* losses it is not one quantity but several: the share of QLIKE damage repaired and the share of squared-error damage repaired are different estimands, and when this paper reports all three and finds them close, that is three estimands agreeing rather than one quantity measured three ways. The agreement is reported as robustness, not as identity.

$R(\delta)$ is a share of forecast loss and not an amount of money. When this paper says a trade can be *priced*, it means that the two things being compared can be put on one ruler and that the ruler has a zero and a natural restoration reference point at 100%: a data feed and an options subscription can be ranked against each other, a rate near zero says the purchase is worthless, and a rate above 100% says the substitute supplies information the fresh domestic-history model never had. The reference point is not a mathematical ceiling on $R(\delta)$, and Section 6 reports values above it. Converting that into a currency needs a decision problem the holder supplies, a value at risk or a capital buffer or a volatility target, and Section 11 names it as the obvious next study rather than guessing at one here. What the paper does offer in a practitioner's own units is the clock of Section 9: the same recovery read as the age of a mark.

One caution, which Section 7 will need: the measure assumes the substitute is *restoring* something delay destroyed. A source that also carries information the domestic history never had can exceed 100%, and when it does, the ratio has stopped answering the question it was built for.

The denominator can fail in the other direction too: where delay costs a forecaster little, $S_{own}(0) - S_{own}(\delta)$ approaches zero and the ratio is weakly identified, so a uniformly valid confidence procedure must be free to return an unbounded set (Dufour 1997). Every interval this paper reports for a rate on the daily equity panel inverts a HAC t-test after Fieller (1954), which is free to return a bounded interval, a union of two half-lines or the whole line according to what the data support; the monthly and housing measurements are resampled instead, for the reason Section 4.6 gives, and are the only rates that are. A percentile bootstrap of the same ratio is free to do none of that: it reports finite endpoints whether or not they mean anything, and Section 4.6 measures what that costs in coverage and in width. Those figures are properties of that generator and not guarantees about the

observed loss process, and the same qualification applies to every coverage figure quoted in this paper.

The bootstrap is kept beside the inversion in the replication output, because two implementations of one quantity agreeing catches arithmetic errors, and it is reported in the tables only where the quantity is not a ratio of two means. Section 4.6 gives a construction that remains valid there, and Section S5 applies it to the state where the risk is greatest. On this sample the denominator never gets close enough for the pathology to bite, but it is a property of the measure itself, and anyone reusing $\mathrm{R}(\delta)$ on a series where delay does little damage should check it rather than assume it.

What the rate is at the far end of the grid. The denominator is $S_{\mathrm{own}}(0) - S_{\mathrm{own}}(\delta)$, and by eleven weeks the stale domestic model has almost nothing left: $S_{\mathrm{own}}(55) = 0.0072$ against $S_{\mathrm{own}}(0) = 0.5188$. The denominator is therefore within a hundredth of $S_{\mathrm{own}}(0)$ itself, and $\mathrm{R}(55)$ is, to that accuracy, the cross-sectional model's skill divided by the domestic model's undelayed skill rather than a share of anything the delay took. The two readings coincide here because the delay took almost everything, which is a fact about this series and not a property of the measure. The share interpretation is cleanest in the middle of the delay grid, where $S_{\mathrm{own}}(\delta)$ is materially positive and numerator and denominator are genuinely different quantities; a reader carrying $\mathrm{R}(55)$ should carry it as the ratio of two levels it has become. That plateau has a specification behind it rather than a curve fitted to it, and Section S26 states it and tests the one implication that is not tied to the endpoint it predicts.

4.6 Inference for a ratio whose denominator can vanish

The substitution rate is a ratio, and Section S3 reports what inverting a HAC t-test after Fieller does to the intervals a bootstrap reports: it leaves every one of them an ordinary interval rather than an unbounded set, and it widens them, by up to 8.2 percentage points. Fieller's theorem (1954) inverts a t-test on the linear combination $\mathrm{num} - \mathrm{R}\cdot\mathrm{den}$, whose expectation is zero at the true rate whatever the denominator is, and solves the resulting quadratic for the set of rates not rejected. Writing $\bar{d}_1$ and $\bar{d}_2$ for the two sample means of (10), $\hat{V}$ for their HAC covariance matrix and z for the critical value, squaring the t-test leaves the set $\{R : AR^2 + BR + C \le 0\}$ with

$$A = \bar{d}_2^{\,2} - z^2 \hat{V}_{22}, \quad B = -2\,(\,\bar{d}_1 \bar{d}_2 - z^2 \hat{V}_{12}\,), \quad C = \bar{d}_1^{\,2} - z^2 \hat{V}_{11}. \tag{12}$$

When the denominator is well separated from zero the solution is an ordinary interval close to the delta-method one. When it is not, A changes sign and the confidence set is correctly a union of two half-lines, or the whole line, which states that the data do not pin the ratio down and is a statement a percentile bootstrap cannot make.

The sign of A has an exact form: $A > 0$ if and only if $|\tau| > z$, where $\tau = \bar{d}_2 \div \sqrt{\hat{V}_{22}}$ is the denominator's own HAC t-statistic. The interval for the share of the damage repaired is bounded exactly when the damage is itself significant at the same level, and at the critical value this paper uses that boundary sits at $\tau = 2.11$. The two ratios (12) inverts do not share that denominator, which is why a calm market can carry a high τ on one and a weakly identified rate on the other; Section S2 gives both. Under weak identification a uniformly valid confidence procedure must permit unbounded sets (Dufour 1997), and the asymptotic Fieller-type inversion used here is one standard construction with that property rather than the only one. It is asymptotic, since it uses an estimated HAC covariance, and the critical values it is inverted at are not normal ones, which is what the last part of this section is about.

The variances must be HAC, with the same Bartlett kernel used in Section S2, because the five-day targets overlap. Independence-based variances are not merely conservative here: substituting them returns intervals materially *narrower* than the block bootstrap at every delay, against loss differentials measured to carry positive serial dependence out to lag 38. Overlap invalidates independence-based inference generally; whether iid standard errors come out smaller depends on the signs and magnitudes of the serial covariances, so the narrowing was evidence rather than proof. And $\delta = 0$ is absent from the comparison throughout, because there the denominator is identically zero by construction, the one delay at which the ratio does not exist at all, rather than being merely hard to pin down.

The truncation length is the third. A delay-δ loss differential compares a forecast that is δ days stale with one that is current, so any two evaluation dates fewer than δ days apart share stale information by construction, and the dependence runs past the target's five-day overlap by the delay itself. No single bandwidth settles how far past, and under normal critical values none attains the nominal level. The reason is the taper rather than the length: the Bartlett kernel weights the covariance at lag h by one minus h over the bandwidth and so discards a share of the long-run variance at every finite bandwidth, and a fraction measured against the true long-run variance cannot reach one at any length. What is left once the taper is priced is not a length. It is a normal quantile applied to a statistic whose long-run variance is estimated rather than known.

The intervals reported here therefore invert the heteroskedasticity- and autocorrelation-robust t-test at a fixed $b = M/n = 0.05$, with the fixed-b critical values of Kiefer and Vogelsang (2005), which price that estimation instead of assuming it away. At $n = 4{,}862$ that is a bandwidth of 243 days and a two-sided critical value of 2.11 rather than 1.96. The limit is pivotal, so the critical value is simulated on independent draws under two conditions: it approaches 1.96 as b approaches zero, and rises with b wherever the difference exceeds the simulation's own noise. At the b used here it is 2.11, with a Monte Carlo standard error near 0.02, and agrees to two decimals with Kiefer and Vogelsang's published approximation at $b = 0.05$. On the generator of Section S2 the inversion covers 94.7%, 94.6% and 94.6% at three identification strengths, each with a Monte Carlo standard error of a third of a point, against 92.5%, 93.0% and 92.5% for the normal-quantile procedure it replaces, at about a tenth more width. Their mean is 94.6% against a nominal 95%, with a standard error of a fifth of a point, so the shortfall is small and is not noise. It is the taper, which fixed-*b* does not price, and it is larger on the sample than in the simulation: this generator's dependence stops well inside the bandwidth, while on the observed loss differentials the same kernel retains 85.6% of the long-run variance, implying 93.0% (Section S3). Which of the three applies to an interval is a property of that interval, and is printed beside every set reported here: the denominator's own t-statistic τ. Section S2 states the strength each table should be read against, and no claim in this paper rests on a set weaker than the middle of the three.

A self-normalised statistic (Shao 2010), which uses no bandwidth at all, gives the same readings with wider sets, and is reported in Section S2.

Two things are disclosed rather than repaired. The first is where the moving-block bootstrap is still used. Three families keep it: the effective age of Section 9, which is an inverse interpolation and has no t-statistic to invert; the difference between two substitution rates, which is not itself a ratio of two means; and the monthly and house price measurements, which are run at a different frequency and on too few observations for $b = 0.05$ to be inherited without recalibration. Section S2 names the tables

in each family and the reason in each case. Every other interval on a substitution rate in this paper and its appendix, including the market-state ladder of Table S7, is a fixed-b Fieller set. The bootstrap intervals that remain cover 90.9% where the denominator is strongly identified and 95.6% where it is weak, at two to three times the width of the set the inversion returns there.

The second is that the bootstrap cannot be used to corroborate the inversion at a matched length, because a moving-block bootstrap at block b destroys dependence past lag b exactly as a Bartlett kernel at bandwidth b does; the two agreeing is a check that two implementations of one quantity agree, and is not evidence about coverage.

5. What breadth recovers

This section applies the design of Section 4 to the eight equity series and reports what a current foreign cross-section returns at each delay, unconditionally and by market state. The test window runs from May 2007 to September 2026, 4,862 days.

Table 2. Out-of-sample skill against a trailing mean of already-resolved outcomes, and the substitution rate under each loss. Skill is the fraction of benchmark loss removed, so the two pairs are comparable. The benchmark is one forecaster on both scales: it forecasts that trailing mean on the log target, and on the variance scale it is exponentiated using its own smearing factor and reconstructed using the median available at the forecast origin. That is not the trailing mean of natural variance, and Section S25 gives the size of the difference. Intervals are the fixed-b Fieller sets of Section 4.6, the same construction the abstract and every other table here use. The rate is a ratio of two means once the shared benchmark loss cancels, so the inversion applies to it directly; the whole-ratio bootstrap of the same differentials is in the replication output beside it and is narrower, which is the direction the coverage study measures. The $\delta = 55$ cell on R^2 is an independent recomputation of the set Section S2 reports and agrees with it. R^2 is on the log scale and is not proxy-robust; QLIKE is. This table is computed on the target-dated definition of Section 4.2, as the equity-result tables are, and at eleven weeks it reads 71% as the abstract's implementable figure does; the two conventions differ by at most a point at any delay.

δ	R^2 own	R^2 cross	$R(\delta)$ on R^2	QLIKE skill own	QLIKE skill cross	$R(\delta)$ on QLIKE
0	0.5188	0.5089	n/a	0.6186	0.6060	n/a
1	0.4738	0.4732	−1% [−28, 30]	0.5719	0.5699	−4% [−48, 47]
3	0.4094	0.4492	36% [16, 56]	0.4995	0.5475	40% [8, 69]
5	0.3637	0.4434	51% [33, 68]	0.4421	0.5416	56% [28, 79]
8	0.3110	0.4346	59% [43, 74]	0.3651	0.5322	66% [40, 84]
13	0.2332	0.4150	64% [48, 77]	0.2258	0.5123	73% [42, 87]
21	0.1478	0.3973	67% [50, 79]	0.0932	0.4972	77% [45, 90]
34	0.0792	0.3857	70% [53, 81]	−0.0121	0.4904	80% [51, 91]
55	0.0072	0.3729	71% [54, 83]	−0.0475	0.4854	80% [52, 93]

Table 2 reads down a column and then across. The first column is what a forecaster loses to delay alone: R^2 on the domestic block falls from 0.5188 at zero delay to 0.0072 at eleven weeks, a loss of 99% of the skill it began with, so by then a stale mark is worth about what a trailing mean of already-resolved outcomes is worth and no more. The second column is the same forecaster with the foreign cross-section added, and it falls far less, from 0.5089 to 0.3729, because the peers are current whatever the domestic delay is. That is still a loss of about a quarter of its skill: the cross-section slows the decay rather than stopping it. The third column is the ratio of the two gaps, which is $R(\delta)$ in (9): 36%

of the delay's damage repaired at three days, 51% at a week, 71% at eleven weeks. The two QLIKE columns repeat the exercise on the proxy-robust loss and reach the same shape with larger numbers, 40% to 80%, so the result is not an artefact of the scale the headline is quoted on.

The two losses do not agree about everything. Under R^2 the stale domestic model is still marginally above the benchmark at the longest delays, 0.0792 at $\delta = 34$ and 0.0072 at $\delta = 55$. Under QLIKE it is below it, -0.0121 and -0.0475. Both statements are about the same forecasts and the same benchmark; what differs is which errors the loss punishes. Squared error on logs is symmetric in the log and so forgiving of a forecast that is too low in variance terms, which is what a stale mark mostly is once volatility has moved; QLIKE punishes exactly that asymmetry, and it is the proxy-robust one. Where the two conflict we give weight to QLIKE, which makes the reading of the long end the harsher one: a mark eleven weeks old is not merely worth about what a constant is worth, it is worth slightly less. The $\delta = 1$ row is the one that does not fit the pattern, and it is the row that should not: at one day the domestic history is barely stale and there is almost nothing for the cross-section to put back, so the rate is indistinguishable from zero under both losses.

One caveat attaches to every figure in that third column, and Section S5 establishes it. An unconditional rate pools three states in which this mechanism behaves quite differently, and the cut is a VIX tercile rather than a two-way split. The tercile boundary was chosen after a ladder of alternative cuts had been run, which Section S5 records; the ladder is therefore reported as a search family, and the state claim below is tested accordingly. That the *effect* is larger under stress is a claim about a difference, and is established on the share of domestic error the cross-section removes rather than on the rate: estimated as one parameter on the full sample, that share is 42.0 points higher in the top 5% of VIX days than in the bottom 67%, with a resampling interval of $[25.2, 55.8]$ and a p-value of at most 0.0002 under a statistic that prices the choice of rung. The same contrast computed on the rate itself is 15.0 points with an interval of $[-7.4, 37.5]$, which spans zero, so the rate's own cells describe the states without separating them.

In stressed markets the rate is 81% at eleven weeks with an interval of $[67, 90]$, and the point estimate is 81% in each half of the sample taken separately. The half-sample sets are too wide to constitute a stability test, and are not reported as one. In the middle tercile it is 53% $[28, 76]$, lower at the point estimate, and its set overlaps the stressed one, as all three do at every delay. In calm markets the same quantity spans more than a hundred points and excludes zero at none of the four delays. It is imprecise rather than unidentified: the denominator's own t-statistic runs 2.9 to 3.5 there, above the $\tau = 2.11$ boundary of Section 4.6, so the Fieller construction returns bounded sets at every calm delay and does not classify the cell as weakly identified. This paper does not establish why the calm estimate is so imprecise.

Table 2's 71% therefore pools a well-determined estimate, a weaker one, and a non-result, and the ordering across the three is what the mechanism predicts. We report it because it comes from the original specification, the one fixed before any alternative was run, and because the literature quotes unconditional rates, but the number a practitioner should carry is the conditional one in Table S4.

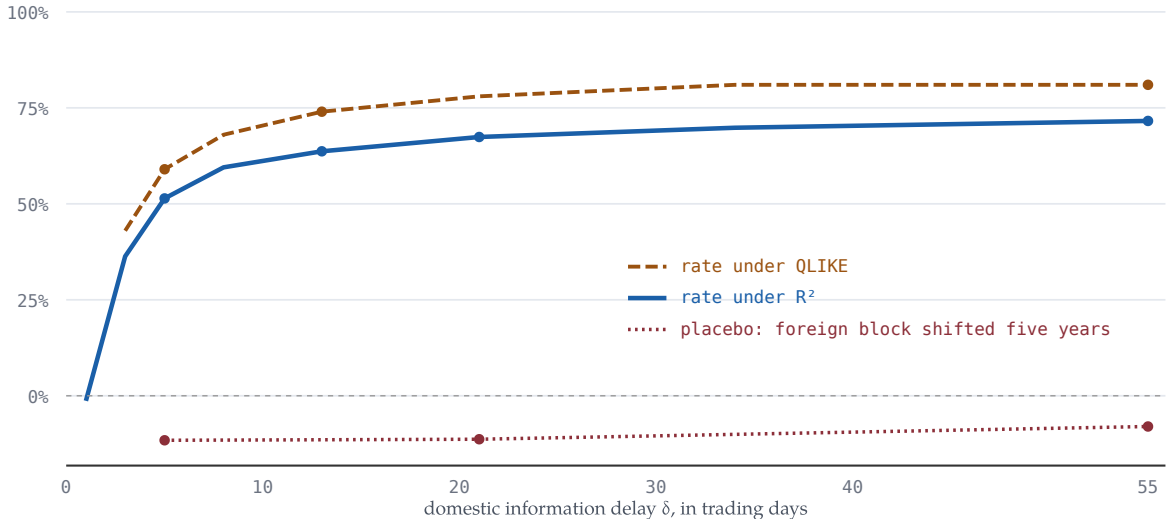


Figure 2. Breadth buys back almost nothing at one day, half the loss within a week, and around three-quarters by eleven weeks. The placebo, described in Section S3, keeps the same foreign series but destroys their alignment with the target, and the effect disappears.

Three things stand out. At one day the rate is indistinguishable from zero, as it should be: the S&P's own recent history is still the best description of the S&P, and the foreign block we selected produces no measurable net improvement on it. By eleven weeks the picture has inverted. The domestic model's out-of-sample R^2 has fallen to 0.0072, so a mark eleven weeks old is worth no more than a trailing mean of outcomes that had already resolved, while the cross-sectional model still holds 0.3729, which is 72% of the 0.5188 a fully current domestic mark delivered. That denominator is the domestic model at zero delay, the best either model does with current data; against the cross-sectional model's own zero-delay figure of 0.5089 the same ratio reads 73%. Between those extremes the rate climbs steeply and then slows, which raises the question of where it stops.

The third loss belongs here because it is the robust one that most resembles the headline. Under squared error on the natural variance scale the substitution rate reads 74% at five days and 80% at eleven weeks, against 51% and 71% on the log scale. Both robust losses therefore put the rate above the headline, and they agree closely: QLIKE also reads 80% at eleven weeks. No interval is quoted for either variance-scale rate, and that is a result rather than a reticence. Both are ratios of two means, so the inversion of Section 4.6 applies, and run on them it declines to identify them: the median-normalised rate returns two half-lines at every delay and the natural-scale rate returns the whole real line at six of the eight. The denominator there is squared error on untransformed variance, which a handful of crisis days dominate, and a ratio whose denominator is estimated that imprecisely cannot be pinned down however it is bracketed (Section S25). That is why QLIKE, not MSE, is the robust loss we lean on. A percentile bootstrap returns finite endpoints at every one of these cells, concealing the identification problem rather than reporting it, which is the strongest reason not to quote one here.

Whether the result is mechanical. The objection is taken in its strongest form. At $\delta = 0$ the gross content of the foreign block is $+0.0027$, interval $[-0.0064, +0.0116]$: the cross-section carries nothing about the S&P's current state that current domestic history does not. Figure 3 then orders the rate by coupling. A sceptic can therefore say the paper shows only that correlated series stand in for a hidden series in proportion to their correlation. Three things separate the measurement from that reading.

The share is not knowable in advance: nothing in a correlation says whether the second series repairs a tenth or nine tenths of what staleness takes from the first, and the values measured here run from below zero to above three quarters. The price is separable from the content, and Table 3 separates it, which is what a buyer needs and a correlation cannot give. And the rate is not a function of coupling alone: Section 7 finds that a current volatility index on the asset itself drives the increment to nothing while leaving coupling untouched.

6. The limits of breadth

The rate flattens near three-quarters instead of approaching one, and the reason can be measured. At very large delay the domestic model retains little usable out-of-sample skill, so the cross-sectional model is working almost entirely from foreign information. Its ceiling is therefore whatever share of the S&P's volatility the rest of the world can see.

We measured that share directly. Regressing the S&P's decision variable on the six foreign closes of the same day – the block of Section 4.3 without Bovespa, which enters a session late and so has no same-day close to regress on – gives $R^2 = 0.695$. Roughly 30% of the S&P's current volatility state is not linearly spanned by those six series in this sample. We resist calling that share genuinely local, because the regression does not show it: a non-linear function of the same closes, or a wider cross-section, could reach some of it. What the number does bound is how much a LINEAR model of THESE markets can see, the ceiling the fitted model is working against. The contemporaneous spanning figure above is the bound this section is entitled to, and it is an in-sample bound on a different quantity from the rate. The gap between that bound and the fitted figure is consistent with the price of estimating parameters instead of knowing them, and the two measurements later in this section put a cost of the right order on it; the design does not establish that estimation is the whole of it.

The cross-section is also nearly one-dimensional. The first principal component of the six foreign volatility series accounts for 66.6% of their variation and the first two for 77.0%; on the seven-column block the models actually carry, with São Paulo at its admissible lag, the same two shares are 64.3% and 73.3%, which is the pair Section 6.1 uses. Six markets are close to one global volatility factor plus idiosyncratic noise, which carries a practical implication: a ninth or a tenth correlated index may add relatively little under this linear specification, because what a linear model of these markets can see is set by how much of the S&P is global; counting more global series does not move it.

That leaves the other half of the sentence above: what it costs to estimate parameters instead of knowing them. It is large enough to deserve a number of its own. Breadth is not free. Seven foreign coefficients must be estimated from a finite window, and the estimation is paid for whether or not the coefficients turn out to matter.

Two measurements agree on the size of the bill. The first varies the number of peers at $\delta = 0$ and reads off the slope: the penalty is a straight line through the origin at -0.00137 of R^2 per market, which accounts for 98% of the variation across peer counts and implies -0.0096 for the full seven. The second replaces the foreign block with AR(1) surrogates fitted to each peer, drawing innovations from a multivariate normal with the peers' own contemporaneous covariance so the surrogate cross-section is as correlated as the real one, while carrying no information about the target at any lag. Averaged over one hundred draws this gives -0.0126 at $\delta = 0$, with a Monte Carlo standard error of 0.0003, an order of magnitude below the sampling interval that follows and therefore not the

uncertainty that binds. The two methods differ by 0.003, which agrees on the magnitude without being a formal match, and we do not read a third decimal place from either. About one point of R^2 is the price of carrying the cross-section. Section S21 measures it a third time, with a generator that fits nothing at all, and gets the same answer.

Two things follow. The first settles how the negative increment at zero delay should be read. At $\delta = 0$ the measured difference is -0.0099 and the cost is -0.0126, so the gross content of the foreign block is $+0.0027$. Gross is a difference of two estimated quantities, so we bootstrap it, resampling test days in moving blocks and recomputing the measured difference and the surrogate cost on the same days so that their errors stay correlated. The interval is $[-0.0064, +0.0116]$: indistinguishable from zero, and bounded above at about one point of R^2. This design detects no information in foreign markets about the current state of the S&P beyond what current domestic history already carries, and bounds any such content above at about one point of R^2. Whatever they know about it that fully current domestic data does not already carry is too small for this design to detect, and the negative sign was the bill, not the information.

The second is that the bill grows with delay, from -0.013 at $\delta = 0$ to -0.051 at $\delta = 21$. This is a property of the fitting procedure: as domestic information goes stale the fit weakens, the validation step selects a looser penalty, and the seven regressors are given more rope. Separating the two makes the gross and net readings diverge in a way that matters for anyone deciding whether to build this.

Table 3. What the cross-section knows, and what a forecaster keeps after paying to estimate it. Cost from AR(1) surrogates, one hundred draws, innovations carrying the peers' own cross-market covariance. Gross is bootstrapped in the text: indistinguishable from zero at $\delta = 0$, marginal at $\delta = 1$, positive from $\delta = 2$.

δ	0	1	2	3	5	8	13	21
Net ΔR^2 (measured)	-0.0099	-0.0005	$+0.0179$	$+0.0398$	$+0.0797$	$+0.1237$	$+0.1818$	$+0.2495$
Cost of breadth	-0.0126	-0.0149	-0.0173	-0.0193	-0.0235	-0.0294	-0.0392	-0.0514
Gross ΔR^2	$+0.0027$	$+0.0143$	$+0.0352$	$+0.0591$	$+0.1031$	$+0.1530$	$+0.2210$	$+0.3009$

Gross content clears zero at one day, $[+0.0004, +0.0279]$, by a margin of 0.0003 at the lower endpoint, and is clear of it with room to spare at two, $[+0.0164, +0.0533]$, which is where the net figure crosses as well. That one-day margin is close enough to the boundary that we report it and decline to build on it. A band in which the foreign block holds something real that cannot yet pay for its own estimation is therefore consistent with these data at one day, and no more than that. At $\delta = 21$ estimation consumes 17% of what the cross-section knows, $+0.3009$ gross against $+0.2495$ net. That gap is not a law of nature. It is the cost of a particular estimator, and a cheaper one should recover part of it: a factor representation of the foreign block instead of seven free coefficients, which the principal-component result above already suggests would lose little. Section 6.1 fits that compressed representation directly.

6.1 Fitting the compressed representation directly

Two facts above sit next to each other without being joined. The foreign block is nearly one-dimensional, its first principal component accounting for 64.3% of the variation and its first two for 73.3%. Carrying seven regressors costs about a point of R^2 at $\delta = 0$ and more as the delay grows, measured by replacing the block with surrogates that carry no signal. It is natural to conclude that a

cheaper parameterisation would recover part of that cost. That is an inference from a hypothetical generator, not a demonstration, and the demonstration is cheap.

The block is compressed and compared. The eigenvectors are recomputed at every refit from the covariance of the training window alone, on columns standardised with training statistics alone, with each eigenvector's sign pinned so the factor does not flip arbitrarily between refits. A basis computed once on the whole sample would leak: “the first component of the foreign block” is not something a forecaster in 2009 could have written down from 2026 data. An equal-weighted mean of the seven series is included as the compression that requires no estimation at all.

Table 4. Out-of-sample R^2 for four representations of the same seven foreign series, against the domestic-only model. PC1 and PC2 use one and two real-time global factors; FULL is the seven-coefficient model the rest of the paper uses. The last columns test PC1 against FULL, which are non-nested, by Giacomini-White.

δ	domestic	+ mean	+ PC1	+ PC2	+ all seven	PC1 − all seven	GW z
0	0.5188	0.5199	0.5200	0.5176	0.5089	+0.0111	2.62
3	0.4094	0.4597	0.4609	0.4584	0.4492	+0.0117	1.79
5	0.3637	0.4536	0.4551	0.4529	0.4434	+0.0117	1.72
13	0.2332	0.4244	0.4269	0.4252	0.4150	+0.0120	1.53
21	0.1478	0.4086	0.4117	0.4089	0.3973	+0.0143	1.56
55	0.0072	0.3873	0.3918	0.3889	0.3729	+0.0189	1.44

One real-time factor beats all seven series at every delay, by 0.011 to 0.019 of R^2, and the margin is larger at four of the five steps along the delay grid, in the direction the surrogate cost moves without tracking it step for step. One qualification belongs with that, and it is the one a reader of Table 4 will reach for: the gain clears conventional significance at $\delta = 0$ alone, where the Giacomini-White statistic is 2.62, and at no other delay. The statistic falls along the grid to 1.44 at eleven weeks, which is the weakest cell in the table rather than a second significant one; the decline is not strictly monotone, since the thirteen-day cell at 1.53 sits three hundredths below the twenty-one-day cell at 1.56. The table establishes a consistent sign at six of six delays and a magnitude of about one point of R^2, not a significant gain at the long delays; we lean on the pattern and on the $\delta = 0$ cell, and the recommendation of Section S30 rests on the size of the gain rather than its significance. The real-time first component explains 66.6% to 66.7% of the block's variance averaged over refits, against 64.3% in the full sample, a training window being the more concentrated of the two, which says the basis is stable enough to be estimated as you go.

Reporting a new preferred specification on the log scale alone would be inconsistent with Section S25, which exists to establish that the scale changes the answer. Under QLIKE, the proxy-robust loss, one factor wins at every delay, exactly as it does under R^2. Under squared error on the variance scale a compression wins at five of six delays; the exception is $\delta = 0$, where the seven-regressor model scores lowest. That loss is dominated by a handful of crisis days, and Section S25 reports that neither variance-scale rate is identified at most delays, so the exception is a statement about tails. Where the three losses appear to disagree elsewhere they are choosing between one factor and two, which differ by thousandths and are not a disagreement about anything.

The compression also raises the substitution rate. Recomputed with the same fixed-b inversion as the headline, $R(55)$ is 71.5% [54.3, 83.4] for the seven-regressor model, which is Table 2's 71% carried to

one decimal so the compressed models can be read against it on the same scale, 74.3% [59.1, 84.1] for the equal-weighted mean and 75.2% [60.1, 84.9] for one factor. All three are the construction of Section 4.6. Every compressed estimate sits inside the interval the seven-regressor model already carries, so the two are not distinguishable and we do not replace the headline with the larger figure. We report 71% as the original specification's full-block estimate, note the compressed estimate beside it, and state that the difference is inside the noise of both.

That leaves four specifications doing four jobs, and naming them settles which number answers which question. The equal-weighted mean is the *practical* model: it needs no estimated eigenvector, it is the cheapest thing a desk can build, and it recovers 74.3%. The seven-regressor block is the *original benchmark*, kept because every conditional and robustness result in this paper is computed inside it, and it recovers 71%. The real-time factor is a *mechanism test* rather than a recommendation, because it locates the recovery in one component. And the state-space filter of Section S20 is a *structural alternative*, reported and not adopted. The paper's headline is therefore the conservative member of that set, and a practitioner following Section S30 should expect the 74% to 75% band rather than the 71% the abstract quotes. That gap is a property of the paper's own caution.

Two qualifications, and the first is the more important. The equal-weighted mean captures nearly all of the gain: PC1 beats it, but by 0.0001 of R^2 at zero delay and by at most 0.0045 at any delay. What is being bought is the decision to compress, not the machinery of compressing, and a practitioner should take the average and skip the eigenvector. Second, this changes the $\delta = 0$ row. The seven-coefficient model scores 0.5089 against the domestic model's 0.5188, the small negative this paper has declined to interpret, while PC1 scores 0.5200, above it. We do not claim the cross-section helps at $\delta = 0$ on that basis: it is a difference of 0.0012 with no interval attached, and a difference carrying no interval is not a finding whichever way it points. What the row supports is narrower and sufficient, that the negative at $\delta = 0$ belongs to the parameterisation and not to the information.

Both measurements of that bill are relative to a generator we chose, and the surrogate one preserves first-order persistence and contemporaneous covariance and nothing else: not heavy tails, not volatility-of-volatility, not crisis clustering. Section S21 measures the bill again with a generator that fits nothing at all, the real foreign block moved circularly through the sample so that every marginal and every cluster is the real one and only the alignment with the target is destroyed. The unfitted generator returns the fitted generator's number, so the objection does not survive the test, and Section S21 records the two ways we got that shift wrong before getting it right.

The recovery lives entirely in the leading component of the foreign block, with the six orthogonal components worth less than nothing. Section S27 gives that decomposition.

7. The race against implied volatility

The objection this section answers is the one Section 2 names as unavoidable: if a forecaster whose domestic data is δ days stale can read the options market while the mark goes cold, why would they need Tokyo? The headline specification does not in fact use today's VIX: its printed close is stamped after the forecast origin, so the admissible arm carries the previous day's close and the same-day arm is reported only as a sensitivity. Implied volatility is forward-looking and is never stale in the sense this paper imposes, though it carries its own timestamp, tenor and risk-premium content, all three of which are priced below and in Section S28; the work Section 2 cites attributes much of what looks like a cross-market effect to it. So the two candidates go into the same walk-forward, on the same labels,

with the same estimator, and Section S28 of the Internet Appendix gives the race in full. Three results come out of it and the paper depends on all three.

Implied volatility dominates the cross-section. A forecaster holding VIX and VDAX-NEW loses almost nothing to delay at all: R^2 falls from 0.5545 at zero delay to 0.5261 at eleven weeks, against 0.5138 to 0.0009 for the domestic block alone on the same 4,525-day window. Table 2's 0.5188 and 0.0072 are the same quantities on the longer window this section cannot use, and the two must not be mixed. Where the cross-section repairs 71% of the delay's damage, the substitution rate for the options block exceeds 100%. Section 4.5 marks that as the point at which the ratio stops measuring repair: implied volatility is not restoring what staleness took, it is supplying something the realised-variance history never contained. VIX is a risk-neutral expectation of S&P variance over the coming month, priced continuously by people who are not waiting for anyone's mark, and a stale forecaster who can read it is not stale about the quantity they care about.

Given implied volatility, the foreign block adds nothing this sample can detect. The increment from adding the seven foreign series on top of the options block is negative at every delay, and Section 6's estimation-cost argument explains the sign: seven coefficients cost one to five points of R^2 on this sample, and that bill is of a size that accounts for the negative sign. What the evidence establishes is a bound rather than an equality: the widest of the four intervals rules out a gross gain larger than 0.022 of R^2, and leaves anything smaller open. Replacing the foreign series with AR(1) surrogates recovers the same penalty from a block that carries no information by construction, so what looks like harm is the price of the parameters rather than a property of the peers. Section S28 makes the point sharper by adding *one* compressed regressor instead of seven, which removes the apparent harm and reveals nothing underneath it.

Net of the estimation cost the block is significantly negative; gross of it the block's content is not distinguishable from zero, and the least favourable of the four intervals Section S28 reports, $[-0.007, +0.022]$ at $\delta = 55$, rules out a gross gain above about 0.022 of R^2 while leaving anything smaller open. “Redundant” is used in this paper to mean exactly that and never to mean equal: a forecaster holding the options market gains nothing this sample can detect by adding seven foreign closes, and pays for them.

The scope condition is itself a result. Where a liquid options market exists on the asset itself, breadth is redundant in the bounded sense defined above, and a holder should prefer own-asset implied volatility for this forecasting objective, at the tenor matching their horizon. The paper states no price for either feed and no objective beyond forecast accuracy, so it stops short of a purchase recommendation.

The scope of that conclusion. It was drawn on two assets. The S&P and the DAX are the only targets for which this paper held an own-index volatility index, and the six targets it sets them against are **not** markets without options. Options trade on every one of them, and four of them (the Nikkei 225, the S&P/ASX 200, the Hang Seng and the Nifty 50) each have a published volatility index, which Table S11 records against each of them, and which Section S40 uses to run the comparison within each of four targets rather than across targets. The contrast of Section S13 is therefore between holding your own options market and holding somebody else's, never between a market with options and a market without. On the second reading the scope condition would disqualify the paper from almost every liquid asset, which is not what it says.

Section S40 closes that gap by holding four of those volatility indices (the Nikkei Stock Average Volatility Index, India VIX, the HSI Volatility Index and the S&P/ASX 200 VIX) and running each target twice: once against foreign implied volatility and once against its own, on the same window, the same cross-section and the same estimator, so that coupling cannot move between the two arms because the target does not change. Breadth is worth less once the options are the asset's own on four of four targets, and on four of four again in the strict admissibility arm. The scope condition now rests on six markets rather than two.

What that test does not do is separate options from coupling, and this paper will not claim that it does. Across the eight targets of Section S13 the two are perfectly confounded: every optioned target is more strongly coupled than every unoptioned one. The paired design removes the confound *between* targets and still cannot resolve it, because on those four markets the foreign cross-section adds nothing *either way*: given foreign implied volatility it is insignificant on four of four, and given the target's own it is insignificant on three and significantly *negative* on the fourth, the Nikkei, where it costs 0.0427. There is no positive increment to collapse, and the one cell that moves does so in the direction that makes breadth worse rather than better. Settling it would need a market on which breadth genuinely helps a forecaster holding foreign implied volatility, and in this panel those are the frontier markets of Section S14, which have no volatility index of their own to hold. The limitation is structural, and more data of this kind does not remove it.

The result worth carrying out of this section is accordingly not the difference but the level. At eleven weeks a forecaster holding the target's own volatility index scores 0.3191 against 0.0749 for one holding the foreign block on the Nikkei, 0.4748 against 0.1766 on the Hang Seng, 0.4576 against 0.2583 on the Nifty and 0.4220 against 0.3199 on the ASX, which is 1.3 to 4.3 times as high, and higher on every target at every delay. Where an options market is written on the thing being forecast, nothing else measured in this paper comes close to it. When an own-asset implied-volatility index is published, it is the strongest forecasting signal examined here. That is an information-choice result and not a recommendation to buy an options feed or to trade options, because this paper models neither prices nor utility.

All of which is a conclusion about assets whose truth is observable, and the property that makes a truth observable is very often the same one that brings an option chain with it. Section 8 is where that stops being a limitation and starts being testable, because the case the paper opens with is an asset with no chain at all.

Two qualifications belong with the result and both are in Section S28. The specification here lags VIX a full day, which discards information the live index carried, because its printed close is stamped fifteen minutes after the S&P's and is therefore inadmissible under Section 3's own rule; the same-day close moves the race further towards options, so the admissible choice is the conservative one. And VIX is a thirty-calendar-day index racing a five-trading-day target, which are not the same unit. Thirty calendar days is about twenty-one trading days, so the instrument is measured over a horizon about four times the length of the thing it forecasts, calendar days having first been compared with trading days. Cboe publishes a nearer nine-day index, near-matched rather than exactly matched since nine calendar days is about six trading days, and Section S24 runs the race again with it: the horizon moves the numbers materially and the conclusion of this section not at all.

8. External validity: appraisal-like markets

Everything to this point is measured on the S&P 500, and Section 7 has just established that this is the one asset class where the paper's own answer says breadth does not matter. The case Section 1 opens with is the opposite one: a quarterly-marked private book, an appraised property portfolio, a fund reporting a lagged net asset value. Nobody writes options on those. If the substitution rate is a fact about liquid equity indices then the paper has measured something real and recommended it to the wrong reader, and that is the largest thing that could be wrong with it.

So it is measured rather than argued. Section S31 of the Internet Appendix runs the whole design on twenty metropolitan Case-Shiller house price indices, monthly from 1987 to 2026: smoothed by construction, no intraday range, no option chain on any of them, and a genuine cross-section of twenty marks published on the same day. Delay is not the two-month publication lag every metro shares, which is common and therefore cancels; it is the same quantity as everywhere else in this paper, the amount by which one owner's mark is staler than the cross-section it sits in. An owner whose book is appraised on a cycle running behind the published index for the others is the concrete case, and three months is one quarterly appraisal cycle.

Two coordinates are carried separately, because a monthly index has no intraday range and Yang-Zhang cannot be computed on it. One is next month's log return, which is the quantity an owner cares about and is not this paper's target. The other is the log of trailing twelve-month realised variance over its own five-year median, which is *backward-looking* where this paper's target is forward-looking: forecasting next month's value of it means forecasting something that overlaps the current window in eleven of twelve months, so an own-only R^2 near 0.95 is arithmetic rather than skill, and the small denominator that follows is why the rates on that coordinate are lower. The return target is the closer analogue.

At one quarterly appraisal cycle of extra staleness the rate is **67.9% [43, 91]** on the return target, positive on all fourteen metros with a full sample, and 18.8% [6, 31] on the backward-looking variance coordinate, positive on thirteen of fourteen. Both exclude zero. Both sets are monthly block bootstraps rather than the fixed-b Fieller sets used elsewhere, for the reason given in Section 4.6. The two are not equally robust. A monthly return label spans four closing months and the trailing twelve-month variance label fifteen, against a training cut $\delta + 1$ months before the origin: at this delay the return labels share no closing month with the label being forecast and the variance labels share about ten. Section S31 slides the cut back until they share none, at constant window length. The return coordinate is the control and holds, 67.9% to 65.2% with 14 of 14 still positive; the variance coordinate falls to 8.0% with eight of fourteen positive. Its sign survives the overlap and its magnitude does not, so the external-validity claim rests on the return coordinate and on the direction. The per-metro counts are the most favourable reading available rather than the evidence: the twenty metros share one national housing cycle, so fourteen of fourteen is not fourteen independent tests and a sign test across them would overstate what the panel establishes. The intervals do not rest on that count. They resample calendar blocks common to all fourteen metros, so the cross-metro dependence is inside the resampling unit rather than assumed away, and prices the one shared cycle the panel has. The shape of Table 2 survives too: at one month the rate cannot be told from zero on either coordinate, exactly as it cannot at one day on the S&P, and it climbs from there.

Two objections are answered in Section S31 rather than deferred. The index is a three-month moving average, so consecutive monthly changes share transaction months and a smoother could

manufacture a rate; read every third month on each phase of the quarterly grid, where the artefact explanation predicts zero, the rate is 57% at one quarter of staleness and positive in 35 of 42 metro-phases. That 57% pools each phase's increments before dividing, rather than averaging one ratio per metro as the headline does, because on fourteen metros a single one whose own delay cost is near zero dominates a mean of ratios. The distinction decides how much the check establishes: all three phases exclude zero under the pooled estimator and one of the three does under the mean-of-ratios estimator, so the quarterly grid rejects the moving-average artefact as the explanation without pinning the magnitude down. And these series carry a large shared seasonal, so a fresh peer could be revealing the calendar rather than the market; a placebo peer built to carry the seasonal and nothing else earns 4.2% where the real cross-section earns 73.0%. That is a test which could have convicted the design and did not.

What this settles is narrow and sufficient. The central claim is not a fact about liquid equity indices. It holds on an asset class that is marked rather than traded, that no options market covers, and whose marks are smoothed by the same mechanism that makes appraisal-based valuation slow. What it does not settle is the case itself: nobody holds a Case-Shiller index, and an index is not a book. The argument has moved from an analogy to a measurement one step removed from the asset it is about, which is further than it was and not the whole way.

9. Robustness and scope

The Internet Appendix examines the principal identified threats to each central claim and carries that work in full: eight alternative controls, a multiple-testing adjustment, a temporal split, a leakage check, a second estimator for the inference, and the three scope conditions that bound where the result holds, the option chain, coupling, and the freshness of the foreign feed itself. The third is stated here because every rate in this paper was computed with the foreign block at its freshest admissible value, and a desk whose own foreign feed is a day behind recovers 29% of the delay-induced loss at five days where Table S8 reports 51%, and three days behind recovers none of it. The infrastructure that delays a domestic mark need not be the infrastructure that delivers seven foreign closes the same evening, so the condition is likely to bind on the reader this paper is written for. This section states what those tests returned. It is deliberately short, because a reader who accepts the results does not need the working and a reader who does not should have it in front of them rather than summarised, which is why the working is a companion document rather than a paragraph.

The benchmark was improvable, and improving it costs the rate three to four points. Median normalisation removes the level of a series' own volatility, and a domestic block given that level back is stronger than the paper's at every delay: own-only R^2 rises from 0.0072 to 0.0573 at eleven weeks, and the substitution rate measured inside that arm falls from 71% to 67%, a paired difference of -4.1 points with interval $[-9.8, -0.7]$. The direction is the same at all eight delays and the size barely varies, so this is one result and not two. It belongs here rather than in the appendix because it is the strongest objection to the design: part of what the cross-section appears to restore is a level the normalisation threw away, and a fair reading of the paper has to carry the 67% alongside the 71%. Table 1 reports it as its own row, and Section S4 gives the five alternative blocks and how the strongest was chosen, on own-only skill and never on the rate under test.

The measurement survives its own inference. Giacomini-White on squared loss and on QLIKE, HAC-adjusted with a Bartlett kernel at the delay-dependent bandwidth Section S2 measures, favours

the cross-sectional model at every delay tested: two-sided p below 0.01 on squared loss throughout, and between 0.011 and 0.034 on QLIKE. Clark-West agrees, with a caveat developed in the Internet Appendix that is why it is the second opinion and not the test this paper relies on. A placebo that keeps the foreign series but destroys their alignment with the target, shifting the block five years, returns −11.4% [−31, 3], −11.4% [−31, 2] and −8.1% [−25, 6] at five, twenty-one and fifty-five days. Two things are true of those sets and only the first is the test. Each contains zero, so the placebo cannot be distinguished from no effect, which is what the design should return when there is nothing to find. Each also lies entirely below the headline's [54, 83], reported as description rather than as the test: Section S10 withdraws the reading of non-overlapping marginal sets as a test of a difference, and this paper does not get to use it here and refuse it there. Reality Check and SPA, admitting how many models were tried, leave the headline standing.

The secondary grids are tested as a set rather than cell by cell. Across the five of them, 41 of 198 cells are significant against their own resampled distributions, where a null that resamples evaluation days jointly within each calendar group puts the chance-only count at 10.3 on average with a 95th percentile of 25, for a resampling p-value of 0.0060. Judged against a normal critical value the count is 77 against a null mean of 18.3, which is the rule's own size under this resampling – 9.2% rather than 5% – and not dependence; Section S37 separates them. Of the 41, 25 are cells in which the added block significantly hurts, so what exceeds chance is structure and not benefit. None of these grids was pre-specified and no family-wise control was applied within them, so no single cell carries a claim.

It survives a stronger control than its own. The domestic block this paper uses is not the best one available, so eight alternatives were built and priced as a set, with the choice made on own-only skill and stated before any rate was read. The rate at eleven weeks spans 60.7% to 74.3% across them and reads 67.4% against the strongest, a paired difference of −4.1% [−10.6, −0.3] from the published figure. This paper therefore quotes 67% against the strongest control alongside its own 71%: the same two quantities rounded to whole points in Table S3's arm rather than in this envelope. The two agree to one decimal and differ in their intervals, Table S3's being [−9.8, −0.7] against the one just given, because different scripts compute them on the arms each selects. The headline is not an artefact of a weak comparison, and the Internet Appendix shows what a ninth control would have to do to move it. It also isolates why that control is stronger: the trailing-median normalisation of equation (2) puts eight markets on one scale, making a cross-section possible at all, and hides the level of volatility from every model here. Restoring that level is worth more at long delays than the entire cross-section. The comparability the design buys is paid for in domestic information, and the price rises with the delay.

It is a conditional result, not an average one. Split by market state, the rate is 81% [67, 90] at eleven weeks in this sample's stressed markets, with the same point estimate in each half of the sample on sets too wide to test stability, while in calm markets so little damage is done that neither the ratio nor the effective age can be pinned down. Asked in days instead of shares, on a construction with no denominator that can vanish, the calm cell is 9.1 times as wide relative to its own point as the stressed one.

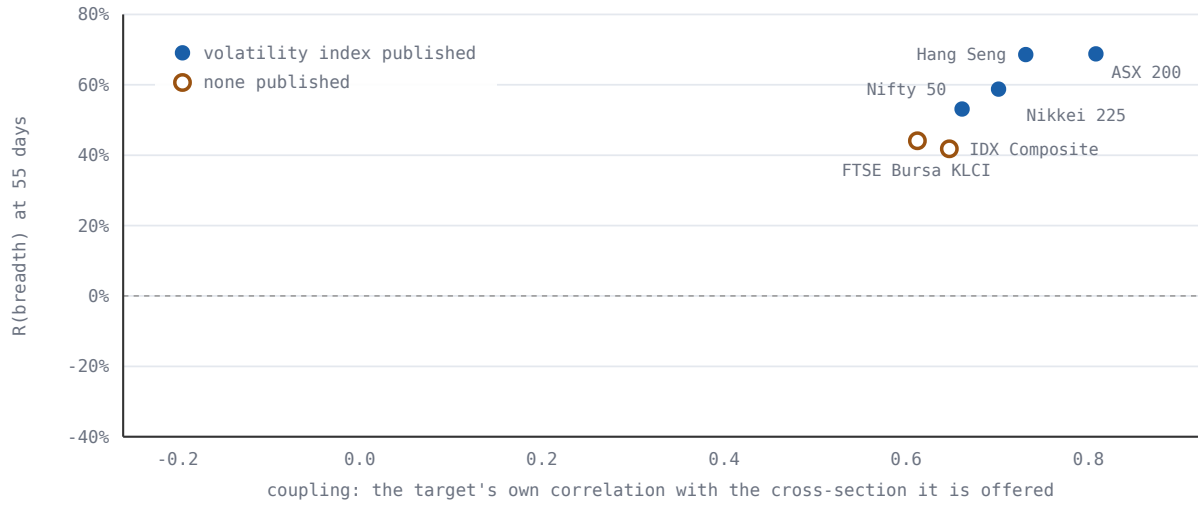


Figure 3. The substitution rate against coupling, for the six targets of Table S11. Coupling is the correlation between a target's own decision variable and the equal-weighted mean of the cross-section it is offered, measured on its own test days; the rate is at $\delta = 55$. Filled markers are targets on which a volatility index is published and hollow markers those on which none is, and the ordering runs through both groups. No line is fitted and the axes are deliberately wider than the data: every target measured here couples between 0.612 and 0.808, so the six points sit in one corner of the frame and the picture of that is the point. What the figure supports is the ordering; what it cannot support is a slope, a zero-crossing or any statement about a weakly coupled market, because this sample contains none. Two further targets were plotted in an earlier vintage and were the only weakly coupled ones; their exports are no longer distributed by the source that supplied them and are not held by us, so a rate computed on them could be reported but never re-run, and this paper reports only what can be re-run. Group membership and coupling remain confounded over the range that is left.

Two conditions bound where it holds, and both can be checked in advance. The first is the option chain, which Sections 7 and 8 establish and this section does not repeat.

The second is coupling, the target's own correlation with the cross-section it is offered. It is a screening statistic and not a calibration rule: it sorts targets into those breadth can help and those it cannot, and it does not return a rate. Across six markets it tracks the substitution rate at +0.92, over couplings running from 0.612 to 0.808, and the figure above is that relation. Six points in one cluster will bear one claim here and not two, and Section S17 separates them. The ordering is robust within the cluster: the rank correlation is +0.943, no single target moves the Pearson figure below +0.896, and the Fisher interval on it, [+0.441, +0.992], excludes zero while being as wide as six points make it. The fitted line is not robust: the S&P/ASX 200 carries a leverage of 0.685 against a mean of 0.333, and removing it moves the slope from +1.469 to +1.972 and the zero-crossing from 0.315 to 0.402 – a crossing that sits below every target in the sample, so the line is extrapolated to it rather than fitted through it. What the sample cannot speak to at all is a weakly coupled market: the two it had are the two whose exports can no longer be obtained, and a rate computed on them could be reported but never re-run.

So the ordering is robust and the crossing is not, which is what the figure's caption says and what that diagnostic licenses: the crossing sits below every target scored here, so it is extrapolated rather than located. A holder whose series is weakly coupled to public markets should expect breadth to do nothing for them, and can find that out from data already in hand rather than after buying a feed; a holder near the crossing should conclude that this paper cannot tell them which side of it they are on.

Three further checks are in the Internet Appendix and are named here so that their absence from this section is not mistaken for their absence from the work: whether the clean lag this paper models resembles the periodic marking a real book undergoes, whether the trailing median leaks the future into the target, and how the rate behaves across five target markets at every delay rather than at one.

10. Limitations

Four limitations are argued at length in Section S33 of the Internet Appendix rather than here: the thinness of the implied-volatility block, two bounds on the tests themselves, and the estimator, which is the largest of them and the one whose direction runs in the paper's disfavour.

The limitations below are individually argued, and a reader who wants the boundary in one place rather than assembled from nine paragraphs should read Table 5 first. It states, for every claim this paper makes, the strength of the evidence behind it and where that evidence is. The four grades are used strictly: *established* means the estimate excludes zero and survives the robustness set, and for the one row that grades a *comparison* rather than a level it means the contrast itself is estimated as one parameter, with a set that excludes zero and goes on excluding it across the robustness set, rather than two marginal sets being observed not to overlap; *supported* means it is consistent across the tests run but rests on a sample too small to resolve alternatives; *suggestive* means the sign is stable and the magnitude is not; and *not claimed* marks a question this paper raises and leaves open. Nothing in the paper should be quoted at a grade stronger than its row.

Table 5. Every claim in the paper, its grade, and where its evidence is. The grades are defined in the paragraph above and are meant to be read as limits rather than as endorsements: a claim graded *supported* is one this paper would not defend against a larger sample.

claim	grade	the evidence, and its limit
A current foreign cross-section returns a large share of what staleness takes, on a liquid equity target	established	71% [54, 83] at eleven weeks, with 67.4% to 75.3% across four specifications, of which the two carrying marginal intervals exclude zero and the two that are paired differences are reported as differences (Sections 5, 7, S4)
The *share of domestic error* the cross-section removes is larger in stressed markets	established	+42.0 points [25.2, 55.8], estimated as one parameter on the full sample from indicator-weighted series rather than compared between two subsamples, with $p \leq 0.0002$ under the maximum standardised contrast across the full ladder, which prices the fact that the ladder's tightening came after a wider version had been run; the earlier reading of this claim rested on two marginal fixed-b sets failing to overlap, and Section S10 withdraws it on both of the usual grounds: non-overlap is not a test of a difference, and the two sets it compared, of 367 and 3,996 days, cover 65% and 94% against 95% on the full window, so they are not two readings of one instrument (Sections 5, S5, S10)
The *substitution rate itself* is larger in stressed markets	not claimed	the same contrast computed on the rate is +15.0 points [−7.4, 37.5] and spans zero; the state cells describe the rate without separating it, which is why the claim above is stated on the share and the point estimates 81% [67, 90] and 53% [28, 76] are reported as description rather than as a difference (Sections 5, S5)
The effect in calm markets	not claimed	the interval spans more than a hundred points and excludes zero at none of the four delays, so no informative estimate is available there; the denominator's own statistic runs 2.9 to 3.5 across the four calm cells and does not classify any of them as weakly identified (Sections 5, S5)
Implied volatility on the asset itself makes breadth redundant, in the bounded sense of Section 7	supported	the foreign block adds nothing detectable once implied volatility is held, at every delay, with a gross gain above 0.022 of R^2 ruled out and anything smaller left open; but two targets only, SPX and DAX, each on the 4,525-day window of Section 7, so the scope is the claim's limit and not its precision (Section 7)
Where the option chain is on another asset, breadth retains incremental value	supported	using the compressed one-regressor peer block and the same pointwise criterion in both groups, the foreign block adds significantly in thirteen of twenty-four cells where only another asset's volatility index is held, and in none of eight where the index is written on the target itself; those cells are in the post-selection audit of Section S37 and few survive its correction, so the pattern is what is supported and no single cell is (Sections S13, S37)
Coupling orders substitution rates across markets	supported	rank correlation +0.943, worst leave-one-out +0.900, and a Fisher interval on the Pearson of [+0.441, +0.992] that excludes zero; six targets, every one of whose exports can be re-obtained, spanning couplings of 0.612 to 0.808, so the ordering is established among well-coupled markets and nothing here speaks to a weakly coupled one (Sections S14, S17)
The fitted coupling line, and the coupling level at which the rate crosses zero	not claimed	six points in one cluster, one of them carrying a leverage of 0.685 against a mean of 0.333; deleting it moves the slope by a third and the zero-crossing by a quarter, and the crossing sits below every coupling measured here (Section S17)

claim	grade	the evidence, and its limit
The measurement carries to an illiquid, appraisal-like asset	supported	57% on non-overlapping quarterly changes, pooling each phase's increments before dividing; all three phases exclude zero under that estimator and one of the three does under the mean-of-metro-ratios estimator the headline uses, 35 of 42 metro-phases positive (Sections 8, S18)
That result is not an artefact of the index's three-month moving average	supported	on the de-smoothed panel breadth still adds at three quarters on 14 of 14 metros; levels carry the reverse filter's noise amplification and only the sign transfers (Section S18)
A current cross-section hurts a forecaster whose own mark is fresh	suggestive	about one point of R^2, negative under all three losses, but the natural-scale MSE cannot distinguish it from zero and no claim in this paper rests on it (Section S9)
What the rate is worth in money	not claimed	$R(\delta)$ is a share of forecast loss; converting it needs a decision problem the holder supplies (Sections 4.5 and 11)
That any of this transfers from a published index to an actual marked book	not claimed	a published index is not a book: no appraisal noise, no manager discretion, and a fixed publication calendar (Sections 1, 8)

The limitation that mattered most through most of this paper was the illiquid case itself, argued rather than demonstrated, and Section 8 narrows it as far as public data allows without closing it. What remains open inside it is stated there: the intervals are wide, the panel's couplings span a band 1.04 times the width of the one that established the coupling relation, so neither panel can test the slope, and an index is not a book. Nobody holds a Case-Shiller index, and a real appraised portfolio would differ from one in ways this panel cannot show. Sections S13 to S15 and S18 of the Internet Appendix carry the four steps by which the gap was narrowed, the reason it is not closed, and the two objections to the narrowing measurement that could have overturned it.

The conditional rate this paper recommends carrying is cut on an instrument the reader it is written for does not have. Both stress definitions rest on VIX, the tercile directly and the calendar route through windows centred on each episode’s VIX peak, and Section 7 is the finding that where a holder can read an own-asset volatility index, that index carries the information instead. On the S&P that is consistent, because the conditioning is a research instrument and the rate is measured in the state rather than gated on knowing it. On the asset class Section 8 reaches, where no such index exists, the conditional figure cannot be computed at all, so what a housing holder can carry is the unconditional rate and the direction of the conditional result rather than its value. Closing this needs the ladder re-cut on a statistic such a holder owns, and that is a measurement this paper has not made.

One limitation applies to every number here and to no single section. All of them were computed with the foreign block at its freshest admissible value, a condition Section 9 states and prices. The measurement is not wrong, but the claim is narrower than it reads: it describes a forecaster with a stale domestic mark and a *current* foreign feed, which is more particular than the holder Section 1 has in mind. Whether an illiquid-asset holder has such a feed is a question about their data vendor, not about volatility, and we cannot answer it.

A second qualifies the headline. The unconditional estimate pools three market states in which the mechanism behaves differently, cut on a VIX tercile whose boundaries are fixed on the whole window: identification is strongest in the stressed tercile, where the rate is large in absolute terms as well as in share; weaker but still resolved in the middle tercile; and weakest in the calm tercile, where it is uninformatively imprecise. Section S5 shows that the imprecision is not an artefact of the ratio, since the calm cell is the wider one asked in days as well. Table 2 pools all three, and the conditional figures in Table S4 are what a reader should carry.

What is left is short. The horizon evidence in Section S23 runs out at one month, where the rate has already fallen to 51% with an interval too wide to lean on; we do not know what happens at a quarter. Equity indices only. Close times are treated as standard, ignoring daylight saving, which moves them by an hour and never reorders the blocks. And the index data are a retail export with no point-in-time guarantee, on which one join cannot be verified by comparison: of the nine two-file exports, eight overlap by 120 to 251 days and agree exactly on every overlapping row, while the S&P's two files abut without overlapping. That is the target series, so contiguity is worth less than an audit, and Cboe's VIX history supplies one: a different vendor on a different data path, measuring the same underlying across the join. Section S19 reports that audit, its three instruments, what size of fault each would have caught, and the one corruption that no instrument here could detect. It also reports something we were not looking for, and that every user of retail market data should know: the retail VIX export carries fifteen rows on dates United States equities were shut. All fifteen fall outside the panel, so no model was ever shown one.

11. Conclusion and further work

What this changes, and for whom. Two results point in opposite directions. Section 7 says that where a liquid option market is written on the asset itself, breadth is redundant on six markets in the bounded sense defined there, and the stronger signal is the published own-index volatility series itself, which scores 1.3 to 4.3 times the foreign cross-section at eleven weeks and higher on every target at every delay. Section 8 says the measurement carries to an asset class no option market covers. Together they bound where the measurement applies. What this paper informs is not the choice facing a liquid equity desk, which already has an own index published on its target, but the one facing a holder of a marked rather than traded book, for whom no such index exists. For that holder the paper supplies a quantity that measures what a candidate source returns against the staleness it cures, a statistic computable beforehand that orders how much any cross-section returns, and a bill of about one point of R^2 per seven estimated coefficients. The rate to carry is the unconditional one, with the direction of the conditional result but not its value. The conditional rate is cut on an instrument the holder of a marked book does not have: where no volatility index is published on the asset, the conditional figure cannot be computed at all.

One scope condition governs everything above and is worth restating where a reader leaves the paper. The delay measured here was imposed, not observed. The S&P 500 does not report late; it was handicapped on a schedule so that the cost of lateness could be measured somewhere the truth is eventually seen, and every rate reported above is a property of that experiment run on real prices rather than of any market's reporting behaviour. The illiquid case the exercise is for is reached in Section 8, and reached only within one step.

The universe here is eight equity indices, and the obvious way to widen it is the one this paper predicts will fail. Section S27 splits the foreign block into its leading real-time component and the six components orthogonal to it, and finds the entire recovery in the first: the remainder is negative at every delay and beats the stale domestic model at none. A ninth and a tenth equity index would supply more of the same factor and more coefficients to estimate, and Section 6 has already priced that trade. Breadth within an asset class is not the frontier. Section S14 says what is: the six targets there differ in coupling from 0.612 to 0.808, and the substitution rate follows that ordering almost exactly. A study that sampled markets ON that axis, rather than by whatever data was to hand, and that reached below a coupling of 0.612, which nothing measured here does, could turn an ordering into a calibration, and a practitioner holding a stale series would then read their own rate off it.

Breadth ACROSS asset classes is, and it is the sharper test because this paper has to make a prediction about it. Sovereign bond futures, the major commodities and the G10 currencies all have observable daily ranges and therefore the same Yang-Zhang decision variable, so the design transfers with nothing changed but the panel. The stress ladder of Section S10 suggests they load on the same global risk factor, and if they do they will be redundant for the reason the orthogonal equity components are, and the mechanism claimed here will have survived a harder test than another equity index. Foreign exchange should be run first, and for no reason about currencies: FX trades continuously, so it has no admissible-timestamp problem at all and is exempt from the staleness penalty of Table S8, which is the largest practical threat this paper has to its own recommendation. Of the candidates here it is the only one current by construction, not by luck of the clock, which is exactly what a holder of a stale mark wants.

Two studies would matter more than any of those, and both are named rather than attempted because each is a paper. The first closes the gap Section 10 leaves open. A Case-Shiller index is not a book: nobody holds one, and the quarterly and de-smoothed tests of Section S18 establish that the result is not an artefact of how that index is built without establishing anything about a real appraised portfolio. What would is a panel of actual marks, whether private equity net asset values, property appraisals or fund marks, carrying a publicly observable peer factor constructed as it stood at each historical marking date, on which the delay curve could be estimated directly. Brown, Ghysels and Gredil (2023) nowcast private equity net asset values from public information and so demonstrate that such a panel can be assembled; what they do not do is sweep the delay, which is the one thing $R(\delta)$ needs.

The second turns the rate into money, and the paper is honest that it currently does not. $R(\delta)$ is a share of forecast loss; a holder wants to know what a data feed is worth. Embedding the forecasts in one decision problem would settle it: a volatility target, a value at risk, or a capital buffer, each of which converts a change in forecast error into a quantity with a currency attached. That would license a sentence this paper cannot write, of the form that a current peer feed reduces risk-measurement error by so much and is therefore worth at most so much a year to a book of a given size. Until someone does that, $R(\delta)$ remains a ratio on one ruler and not an amount of money, which is why Section 4.5 says so where a reader first meets the measure.

Two smaller threads are open and named where they arise: whether the rate keeps rising past one month, where Section S23's evidence runs out with an interval too wide to lean on; and, in the companion note (Vellanikaran 2026), ranking forecasters against a tail target, which needs more independent volatility spikes than nine years of one index contains. The cross-asset panel proposed

above would help with the second as a side effect, since spikes accumulate across markets as well as through time.

Data and code availability

This paper ships with an Internet Appendix carrying the robustness and scope work behind Section 9 in full, including eight alternative controls, the temporal split, the multiple-testing adjustment, the coupling diagnostics, the house price overlap audit and the three scope conditions. Every number in the paper and in that Internet Appendix is printed by a script, and the scripts, the reference output and the source of every document are at `github.com/SunnyAlexV/delayed-information-cross-section`. The repository also carries two machine-readable indexes of the figures quoted in either document: a registry naming, for each figure it covers, the script that produces it and the section it appears in, and a cell-level map from the tables to the lab output behind them. Between them they do not reach every cell of every table, and the repository reports which cells they miss rather than leaving a reader to discover it.

The data are not there. Every vendor behind the inputs prohibits redistributing their files: Investing.com's terms forbid distributing data obtained from the site without written permission, Cboe's permit one downloaded copy for personal non-commercial use and nothing further, Yahoo's forbid reproducing or distributing any portion of their services, and the S&P Cotality Case-Shiller series carry FRED's *Copyrighted: Pre-approval Required* label, which without the copyright holder's permission allows non-commercial educational or personal use alone. Using those series to do research and publishing the results is a different act from republishing the files, and only the second is forbidden; of the thirty-eight input files exactly one, the FHFA index that FRED places in the public domain, could be included and is.

What ships in place of the data is a manifest recording, for every input file, its source, its row count, its date span, the SHA-256 of the exact bytes these numbers were computed from, the digest of the derived series on those dates, and that digest taken one calendar year at a time. A reader who obtains the same exports therefore reproduces every number bitwise, and a reader whose export differs is told so per file and per year rather than left to guess. One source revises rather than appends: FRED restates past months of the Case-Shiller indices, so a later download is a different history and not merely a longer one, and the house price results here are run on the vintage this package pins. Two series that an earlier vintage of this study used are not in that count and are not used anywhere in this paper. The Colombo All-Share and the KSE-100 are no longer distributed by the source that supplied them, and we no longer retain the exports either; only the first is usually what a withdrawal means, and the second is why no confidential channel or vendor permission could produce them. A rate computed on them could have been reported but never re-run, by a reader or by us, so they were dropped and the study re-run without them. The rule is verifiability, it was applied before any rate was re-read, and it would have dropped a target whichever way that target came out. Every number in this paper and its Internet Appendix is therefore computed from an input a reader can obtain.

What the drop costs is stated rather than minimised, because those two were the least coupled of the eight and the only two on which the substitute failed. Removing them removes every observation of a weakly coupled target: the six that remain span couplings of 0.612 to 0.808, a band 0.196 wide, and over a range that narrow no gradient can be assessed. So the scope condition of Section 9 is now an ordering among well-coupled markets and nothing more: it says a better-coupled target recovers

more, and it does not say what happens to a poorly coupled one, because this paper no longer contains a verifiable instance of one. The earlier figures are not quoted anywhere here, since quoting a number we cannot regenerate is the practice this change exists to end.

The repository's README gives the commands: one classifies each input file as identical to ours, the same data with a longer tail, or genuinely different and in which years; another reruns the scripts under a fixed seed and compares each line against the stored output. The house-price scripts cannot be run from obtainable inputs, because they read the twenty-metro Case-Shiller panel, which FRED releases only on pre-approval; a reader without it can regenerate everything else and not those. Nothing here is archived output standing on a digest alone, which was true of an earlier vintage and is the reason the two series behind it were dropped.

The publicly available inputs are cited in the reference list with their repository and series identifiers, as this journal's data-citation policy asks: the Case-Shiller panel and the two national indices, the FHFA all-transactions index, and Cboe's own VIX and VIX9D histories. Every number is computed with NumPy (Harris et al. 2020) and pandas (McKinney 2010) and no other third-party library. The repository's README names which script produces which result. Some scripts were also run on a second machine under a different operating system and matched digit for digit; the rest have been run only on the platform that generated the reference output, and we state that rather than the stronger claim. Nothing is reported that the scripts do not produce.

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Conflict of interest

The author has no financial or institutional interest in the results reported here, and no relationship with any provider of the data used. The author trades Indian equities for his own account. The Internet Appendix includes the Nifty 50 among the eight targets of its coupling result, so the overlap is stated rather than left to be noticed: no position was taken, held or altered on the basis of this work, the paper makes no trading recommendation and reports no tradeable strategy, and nothing in its findings bears on the individual securities concerned.

Disclosure of AI use

During preparation of this manuscript and its Internet Appendix, the author used Claude (Anthropic) as an assistive tool for drafting and revising text and research code.

The research questions, mathematical development, empirical design, data construction, analysis, interpretation of results, and decisions regarding the paper's claims were the author's own. The author independently specified and executed all analyses; reviewed, tested, and validated the code and outputs; and takes sole responsibility for the manuscript's contents, including all methods, results, interpretations, and reported numbers.

No AI system is an author of this work and no AI system held authorial responsibility for any part of it.

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